Institut Polytechnique
Des Sciences Avancées
81 Av. de Grande Bretagne
31300 Toulouse

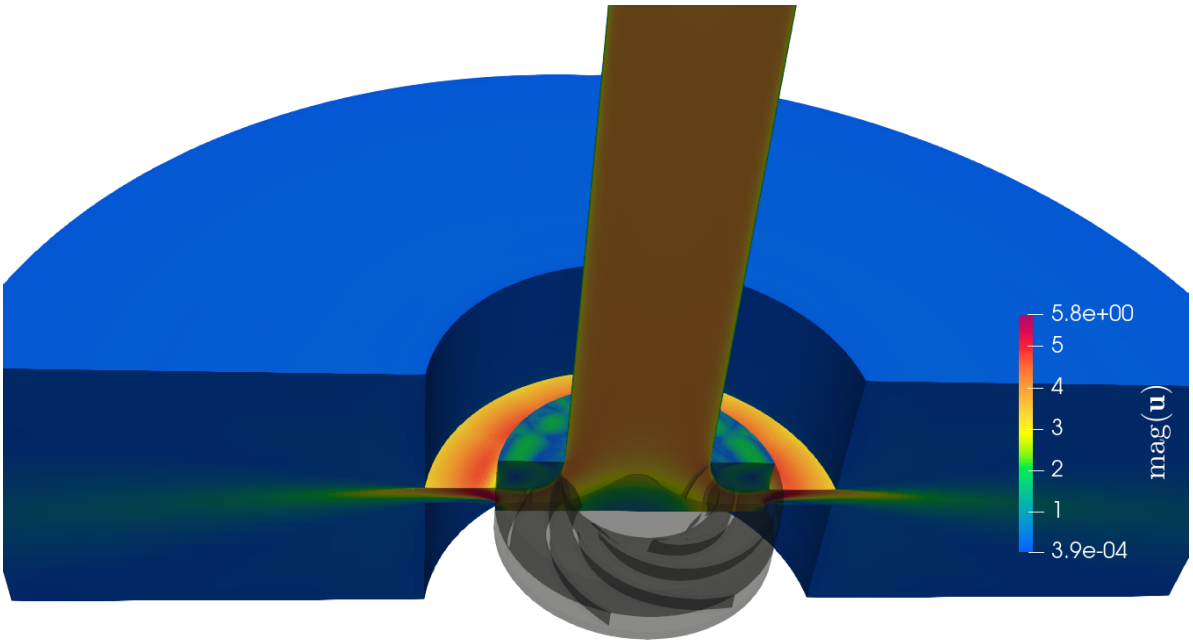
Laboratoire de Mécanique
des Fluides de Lille
Av. Paul Langevin
59650 Villeneuve-d'Ascq

École Nationale Supérieure
d'Arts et Métiers
8 Bd Louis XIV
59000 Lille

End-of-study thesis for engineering degree

Optimization of the performance of a centrifugal pump using Data Assimilation

Intern in Rotating Machines



Cross section of the pump model

Educational tutor: Samy LABSIR
Laboratory tutors: Marcello MELDI, Antoine DAZIN, Miguel MARTINEZ VALERO

Clément THIBAULT - Aéro 5

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Contents

Summary sheet

The aim of this technical assessment is to summarize the gaps and their causes between the course's missions and what has been achieved. It also enables us to suggest what could be done by the next person working on the same subject.

Summary sheet	Clément THIBAULT - Aéro 5 TIE
Internship topic	**Objectives**
Optimization of the performance of a centrifugal pump using Data Assimilation	Employ a DA[1] algorithm to enhance an IBM for the simulation of a centrifugal pump
Main customer	**Used tools**
- LMFL - ENSAM	OpenFoam 9, CONES, VSCode, LaTeX, Git, Python,C++, MPI, Bash, TGCC, Paraview
Conducted studies	
- Coding the DA test case in OpenFoam and CONES - Convergence of the CFD solution (physical and numerical stability/precision) - Convergence of the EnKF, adding tools as inflation	
Results	**Explanation of possible dierences**
- Drop up to 50% of the mathematical NRMSE with DA (almost 100% expected) - Implementation of inflation, in CONES - Implementation of radial expansion, "azimuthal interpolation" in OpenFoam 9 - Discovering of a recirculation bubble in the diffuser	- After the diffuser, the impeller's wake direction is reversed because of the MRF method - The rotating impeller shape combined with the IBM and MRF allows non-physical flow that induce a recirculation bubble - Results of the EnKF are not as good as expected because of the IBM method combined to MRF method, which is not adapted to the centrifugal pump geometry
Encountered diiculties	**Work to be continued**
- Understanding the tools OpenFOAM and CONES - Code everything directly on the TGCC terminal - Initialize physical field of complex instationary simulation	- Use the same type of DA for the analysis of axial compressor internal flows - Synchronize PIV and simulation in the DA - Implement interpolation IBM method

Note: Acronymes are define later. (See glossary)

[1]All acronyms are defined from the next page or available in the glossary.

General Introduction

Since I was young, I've always been fascinated by fluid mechanics. I remember watching feathers, birds and planes flying and being amazed by the way they could fly. I was also interested in the way water flows in rivers and sailing boats. In high school, I began working on plane-related projects and explored them more theoretically through Computation Fluid Dynamics (CFD).

This fascination led me to study physics and mathematics at university, where I discovered applied mathematics including CFD, Artificial Intelligence (AI) and Kalman Filter. During my studies, I've become passionated about applied mathematics and fluid mechanics. That's why I did my last internship at the "Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique" (CERFACS). It convinced me to do a thesis in CFD after "Institut Polytechnique des Sciences Avancées" (IPSA), my current engeneering school.

The subject proposed by the "Laboratoire de Mécanique des Fluides de Lille" (LMFL) really suited me: CFD using an open-source library, coupled with AI and experimental data. So it was with enthusiasm that I started this internship on February 26, 2025, in this laboratory for a duration of six months.

The main objective of this internship is to improve the numerical simulation of a centrifugal pump using experimental data through Data Assimilation (DA). This allow to improve the understanding of the physical phenomena involved in the pump's operation and then to optimize its performance.

The LMFL is a public laboratory with about 40 researchers and 25 PhD students, spread over three sites located in Lille ("École Nationale Nupérieure d'Arts et Métiers" (ENSAM) and "Office National d'Études et de Recherche Aérospatiales" (ONERA)) and on the Villeneuve d'Ascq campus (Centrale Lille, "Centre National de la Recherche Scientifique" (CNRS) and University of Lille).

My main tasks during this internship were:

- Explore and understand the limitations of the baseline Immersed Boundary Methods (IBM);
- Establish the set of sensor signals to be employed as high-fidelity inputs in the DA;
- Ensuring the convergence of the CFD solution using IBM;
- Implementing the DA problem in Open source Field Operation And Manipulation (OpenFOAM) and Coupling OpenFoam with Numerical EnvironmentS (CONES);
- Optimizing the forcing parameters of the IBM using DA with CONES to correct a non-physical behavior.

To present the details of my internship work, the manuscript is structured as follows:

chapter 1 introduces the LMFL laboratory and the ENSAM Lille campus, with a focus on the scientific environment and the social/environmental responsibility of the host institution.

chapter 2 presents the context of the study: first the state of the art and the problem investigation, then the pre-processing of the experimental data, and finally the numerical ingredients employed, including the CFD governing equations, the OpenFOAM solver, the EnKF method and the CONES library.

chapter 3 is dedicated to the application to the centrifugal pump case: the modelling of the pump, the preliminary IBM results, the first simple application of the DA, and finally the analysis of the obtained results.

The report ends with a conclusion, an appendix section gathering additional technical material, the bibliography, and a list of acronyms.

Chapter 1

The LMFL laboratory and Arts et Métiers Lille campus

1.1 Scientific environment and host institution

The LMFL is a public research laboratory jointly supervised by Centrale Lille, Université de Lille, Arts et Métiers ParisTech, ONERA, and the CNRS.

Figure 1.1: LMFL facilities; Top:Villeneuve d'Ascq campus, bringing together Centrale Lille, the CNRS, and Université de Lille. Bottom-left: ONERA. Bottom-right: Arts et Métiers campus (ENSAM) (source : lmfl.cnrs.fr)

Over the past six months, I have been hosted by ENSAM, specifically at the Lille campus. Arts et Métiers is a prestigious engineering school focused on mechanical, industrial, and energy engineering. The Lille campus gives to students a dynamic and innovative environment in a region with a strong industrial heritage.

The LMFL was created on January 1, 2018, from the merger of two research units: the Écoulements Tournants et Turbulents team from the Lille Mechanics Laboratory, and the Expérimentation et Limite de Vol unit. This diversity of expertise enables the research teams to engage in numerous collaborative projects with regional, national and international partners.

Research conducted within the LMFL is organized into three main topics:

- **Turbulence**: Studying and modeling turbulent flows using numerical and experimental methods. The focus is on inhomogeneity and non-stationarity in wall-bounded flows, wakes, jets, and various other flow configurations. This research area includes ten permanent researchers, as well as an equal number of PhD students and postdoctoral researchers.
- **Rotating flows**: Experimental and numerical studies of rotating machines (turbomachines, rotors, propellers) operating under degraded conditions, as well as flow control, cavitation, and multiphase flows, are also key topics. This subject gathers six researchers and about ten PhD and postdoctoral students.
- **Flight dynamics in unsteady and non-uniform environments**: Development of analytical and experimental methods to determine the evolving behaviour of low-speed aircraft in perturbed flight domains. Post-stall dynamics, environmental interactions, and experimental design are among the main research interests. This topic involves about five research engineers and ten PhD and post-doctoral students.

About the LMFL we can also find an interesting document from "Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur" (HCERES). This organization is in charge of evaluating all research structures. Every five years, it publishes a report on the LMFL. The last one[1] was published in February of this year. It presents the laboratory's strengths and weaknesses, as well as its research activities, collaborations, workforce, and future perspectives. It highlights the laboratory's commitment to excellence in research, its multidisciplinary approach, and its contributions to the field of fluid mechanics and related areas. More precisely, the report emphasizes the laboratory's strengths in experimental and numerical methods, its expertise in turbulence and rotating flows, and its contributions to the understanding of complex fluid phenomena.

Some key takeaways from the report:

- The facilities are unique, with wind tunnels and turbomachinery test rigs that allow advanced experimental investigations of external and internal flows.
- It is one of the most active European laboratories in the field of next-generation aircraft design, particularly on electric propulsion systems, supported by high-level experimental setups such as the Eiffel wind tunnel (L1) and a vertical wind tunnel with a rotating balance. The LMFL also leads European research on airship aerodynamics and urban drone flight.
- Its experimental platforms are completed by advanced numerical simulations, with equipment partly acquired through regional "Contrat de Plan État-Région" (CPER) programs. The lab conducts original research on turbulence dissipation mechanisms in non-homogeneous, unsteady flows and on turbulent/non-turbulent interfaces.
- Efforts have been made to create synergy between the lab's three geographically separated sites through transversal research themes in optical metrology, flow control, and data analysis. Weekly LMFL webinars support internal collaboration.
- Despite some staff departures, LMFL has maintained its attractiveness by recruiting top researchers, strengthening historical themes like turbulence and instabilities, and expanding into areas like two-phase aerodynamics and data-driven modeling.

[1] https://www.hceres.fr/sites/default/files/media/publications/rapports_evaluations/pdf/E2026-EV-0590349J-DER-ER-DER-PUR260025134-ST5-LMFL-RF.pdf

- LMFL's scientific output is of high quality, with publications in top journals (e.g., Journal of Fluid Mechanics (JFM)) and participation in major conferences. While flight dynamics publications are fewer, this area is expected to grow through collaborations, particularly with the turbulence team.
- Finally, the lab brings in solid external funding from ERC, ANR, Clean Sky, etc., and works closely with industry partners like CNES and Ariane Group. The leadership was praised for keeping the lab well-organized and for encouraging collaboration across research themes.

1.2 Social and environmental responsibility

ENSAM's Corporate Social Responsibility (CSR), or "Responsabilité Sociétale des Entreprises" (RSE), action plan is structured around five strategic axes[2]:

1. **Strategy, governance and territorial anchoring**: presents ENSAM's actions in terms of sustainable development, in particular the Evolutive Learning Factories 4.0 project aimed at modernizing training and research platforms by fully integrating the CSR dimension;
2. **Education and training**: Training students in eco-responsibility;
3. **Research and innovation**: Ensure that all research themes have direct or indirect impacts on the achievement of international and national sustainable development goals. Ensure free access to research results and data;
4. **Environmental management**: ENSAM aims to reduce its environmental impact by improving energy efficiency, cutting emissions (including Scope 3), and integrating carbon accounting. It also promotes biodiversity and sustainable mobility through staff engagement, responsible purchasing, and green-space management;
5. **Social policy**: Reconciling economic, societal, and environmental dimensions by training staff, teachers, research professors, and stakeholders in sustainable development issues, developing a quality of life policy within the institution, and ensuring equal opportunities in training, research, and human resources. An indicator is the "Qualité de Vie au Travail" (QVT), that is detailed in the "Rapport Social Unique" (RSU)[3] 2024.

The "Comité Social et Économique" (CSE) is the staff representative body within ENSAM. Although specific information on the CSE is limited to the 2022 RSU mentioning the monitoring of the actions of the "Comité d'Hygiène, de Sécurité et des Conditions de Travail National" (CHSCTN) which is the old name of the CSE.

Arts et Métiers also publishes its activity reports, its social balance sheet / single social report and public data then transmitted to the "Commission des Titres d'Ingénieur" (CTI) available online [4]. More specifically, students from Arts et Métiers created a webpage to learn about general CSR and sustainable development: [5] (addressing themes such as ecology, inclusion, gender equality and the fight against discrimination,...).

[2]https://artsetmetiers.fr/fr/ecole-engagee, https://dp-www.s3.ensam.eu/public/2024-11/24_032_plaquette-bilan-RSE_V9_0.pdf
[3]https://dp-www.s3.ensam.eu/public/2025-07/RSU%202024_VCA.pdf
[4]https://artsetmetiers.fr/fr/observatoire-des-donnees
[5]https://artsetmetiers.fr/fr/actualites/developpement-durable-et-rse-faites-le-plein-de-connaissances-sur-savoir

My personal experience of the work environment has been very positive. The working atmosphere within the laboratory is pleasant and conducive to research. The researchers are passionated about their work and share their knowledge with enthusiasm. The team is close-knit and collaborates with researchers from all over the world. The working atmosphere suits me: researchers generally work hard, but there is no pressure regarding working hours. The building is a beautiful old, typical Lille building, but well renovated and adapted to the research work being conducted (both numerical and experimental). The laboratory offers spacious areas, interior courtyards, large lecture halls, and a large workshop for experiments.

Chapter 2

Context of the study

The internship began in February 2025. I was provided with a fix computer, credentials to connect to the "Très Grand Centre de Calcul du CEA" (TGCC) (Irene - Skylake and Rome partitions) and Cassiopée supercomputers, and the link to the Coupling OpenFoam with Numerical EnvironmentS (CONES)'s gitlab[1]. Marcello MELDI, Antoine DAZIN and Miguel MARTINEZ VALERO who are my internship supervisors, showed me around the lab. Then, Tom MOUSSIE and Miguel MARTINEZ VALERO, my co-officies, explained me the basics of Open source Field Operation And Manipulation (OpenFOAM) and CONES, the DA code. I was able to run a few tests on OpenFOAM on the supercomputers, which allowed me to familiarize myself with the tools and the code.

To introduce the topic of my internship, let's start with a simple example: weather forecasting. Meteorologists improve predictions by combining numerical models with real-time observations through a process called data assimilation. This approach corrects model errors and leads to more reliable forecasts.

A similar challenge arises in fluid mechanics, where we aim to understand and predict the behaviour of complex flows. While analytical solutions exist for simple configurations, realistic flows, especially those found in industrial systems, often require numerical simulations. However, high-fidelity simulations are computationally very demanding, making them impractical in many cases.

This difficulty is particularly evident in the case of turbomachinery, such as pumps, which are at the heart of my study. Inside a pump, the flow is highly three-dimensional, naturally unsteady, and involves both fixed and rotating walls. It operates at high Reynolds numbers, leading to turbulence. Moreover, the complex geometry makes the mesh generation challenging, especially near the interfaces between moving and fixed parts. These difficulties motivate the use of techniques like the Immersed Boundary Methods (IBM), which simplifies the treatment of complex geometries while maintaining accurate flow predictions.

Because of these complexities, simplified flow models with adjustable parameters are commonly employed. However, these parameters must be carefully calibrated to capture the underlying physics, typically using experimental data. High-fidelity measurements are thus essential to improve model accuracy.

Manual parameter tuning can work when only a few parameters are involved, but as their number grows, it quickly becomes inefficient and unreliable. This is where mathematical approaches like the Ensemble Kalman Filter (EnKF) become valuable, as they automate parameter estimation by combining experimental data with model predictions. This need for accurate calibration in complex flow configurations is precisely what defines the motivation and objectives of my internship.

[1] https://gitlab.ensam.eu/pe431/cones-dev

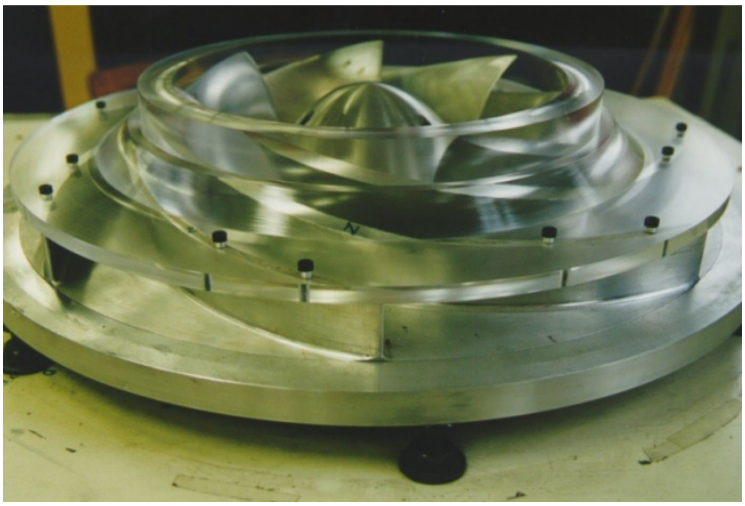
2.1 State of the art and problem investigation

The pump Centrifugal pumps are used since 1475[11] and remain essential in many applications, such as water supply, irrigation, turbomachinery, and industrial processes.

However, despite this apparent simplicity, the internal flow physics is highly complex. Rotor-stator interactions, flow instabilities, pressure fluctuations, and unsteady effects all strongly influence the pump's efficiency and the maximum power that can be extracted[12]. This complexity justifies the need for detailed experimental and numerical studies.

In this work, the studied configuration is a "Société Hydrotechnique de France" (SHF) radial impeller pump, designed for low head applications and high flow rates. The RESEDA benchmark, available at the laboratory, provides a reference case to investigate these complex phenomena under controlled conditions. Its dimensions and rotational speed are chosen to reproduce realistic operating conditions while remaining compatible with experimental diagnostics.

(a) Old centrifugal fan used to move hay (personal picture)



(b) SHF impeller used in the RESEDA benchmark

Figure 2.1: Centrifugal pumps

Different high-fidelity data has been recorded on the diffuser of the pump at the LMFL:

- High-Speed Particle Image Velocimetry (PIV) data obtained by Dazin et al. [4] in 2011;
- pressure data obtained by Fan et al. [5] in 2022.

Both experimental (traditionally) and numerical (increasingly in recent years) approaches have been used to investigate the rotor-stator interactions to improve performance, develop accurate models and permit real-time monitoring and control. Experiences gives us a very precise view of the flow, but only at specific locations. Numerical simulations provide a complete view of the flow, but they are less accurate and require model calibration. Combining both approaches through data assimilation can leverage their respective strengths to achieve a more accurate and comprehensive understanding of the flow inside the pump.

CFD To obtain a numerical simulation of the flow in the pump, a Computation Fluid Dynamics (CFD) can be performed. There are different ways to discretize the space when we do CFD (see Figure 2.2[2]). The most common are structured meshes, unstructured meshes and body-fitted meshes.

[2]https://www.solidworks.com/sw/docs/flow_basis_of_cad_embedded_cfd_whitepaper.pdf

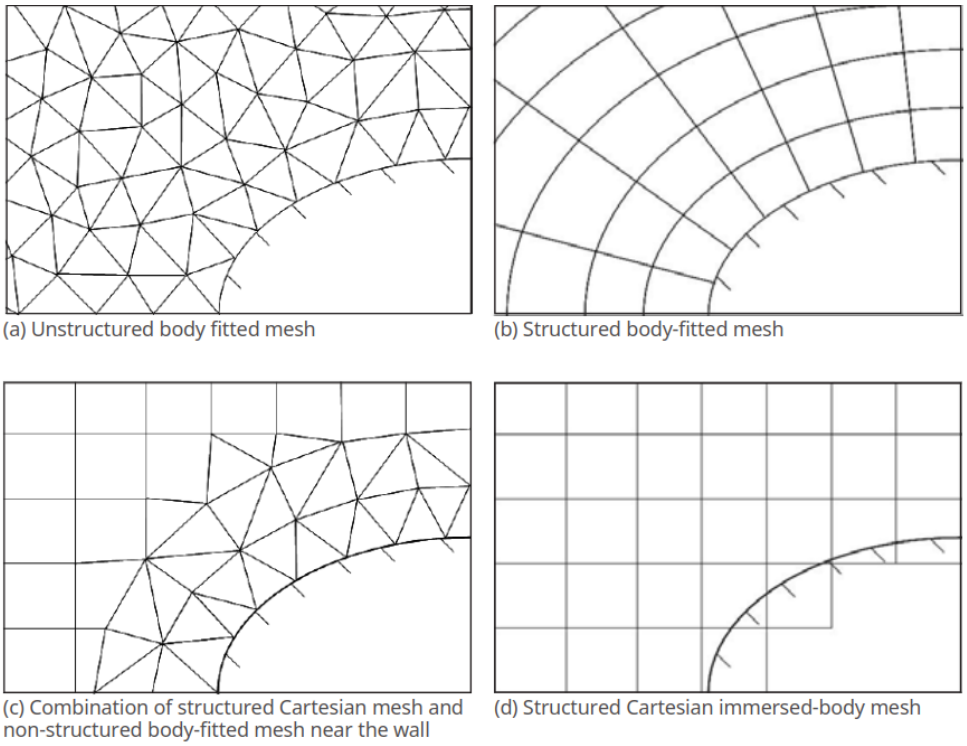


Figure 2.2: Differents strategies for meshing

A structured mesh is a mesh whose connectivity between points can be described in a regular manner, typically using multidimensional arrays (1D, 2D, 3D). In such a mesh, the points are organized in a regular grid, allowing easy navigation through the data. In contrast, an unstructured mesh is a mesh where cells cannot be described in a regular manner. They are defined by arbitrary geometric shapes where the connectivity of the points does not follow a regular pattern. These meshes are often used to model complex geometries where a regular grid would be unsuitable.

The body-fitted meshes are the most accurate but also the most expensive in terms of computational resources in case the grids update at each time step.

IBM is a numerical technique that simulates viscous flow with embedded boundaries on grids that do not conform to those boundaries. This allows to keep numerical efficiency on complex geometries and moving boundaries without the need for body-fitted meshes, which can be computationally expensive and difficult to implement.

In 1972 Charles S. Peskin[9] developped the IBM method. His goal was to reduce the computational time, mainly due to the computation of the mesh refinment based on the Courant-Friedrichs-Lewy (CFL) (see Equation 2.25). He added an IBM forcing term on Navier-Stokes equations, only on the simulated artificial heart valve part. He overcame the numerical instabilitys due to the boundary forces by using an iterative method for solving a fixed-point problem for boundary forces.

In 1999 Angot et al. [2] discovered that using the penalization IMB, the velocity penalized at the wall was the same as the one from a body-fitted approach if $\mathbf{D}$, a forcing parameter defines in the subsection 2.3.1, tend to infinity.

IBM are usually less accurate and need more cells element to keep good precision in the near-wall region, but more efficient in terms of computational resources when the geometry is moving. So it's more and more used in CFD [15]. The maths behind the method are detailed in the Equation 2.27.

Data Assimilation Data Assimilation (DA) is a technique used to improve the accuracy of numerical models by combining them with observational data[3]. It is widely used in various fields, including meteorology, oceanography, and engineering. The goal of DA is to obtain a more accurate representation of the state of a system by integrating information from different physical measurements or hight-fidelity sources.

There exist many methods for data assimilation, distributed in three main categories: **variational**, **statistical** and **hybrid** methods (see Figure 2.3).

- Variational methods (also called classical methods), such as the 3D-Var and 4D-Var, use optimization techniques to find the best estimate of the state of a system by minimizing the difference between model predictions and observations[10];
- Statistical methods (also called sequential methods), such as the Kalman filter and its EnKF variant, update the model state iteratively as new observations become available[3]. They rely on the assumption that the analysis error e^a has zero mean, i.e. $\mathbb{E}[e^a] = 0$, which ensures that the analysis is unbiased.
- Hybrid methods, which combine both variational and statistical approaches. They are not discussed here, as they were not used in this work and fall outside the scope of our study.

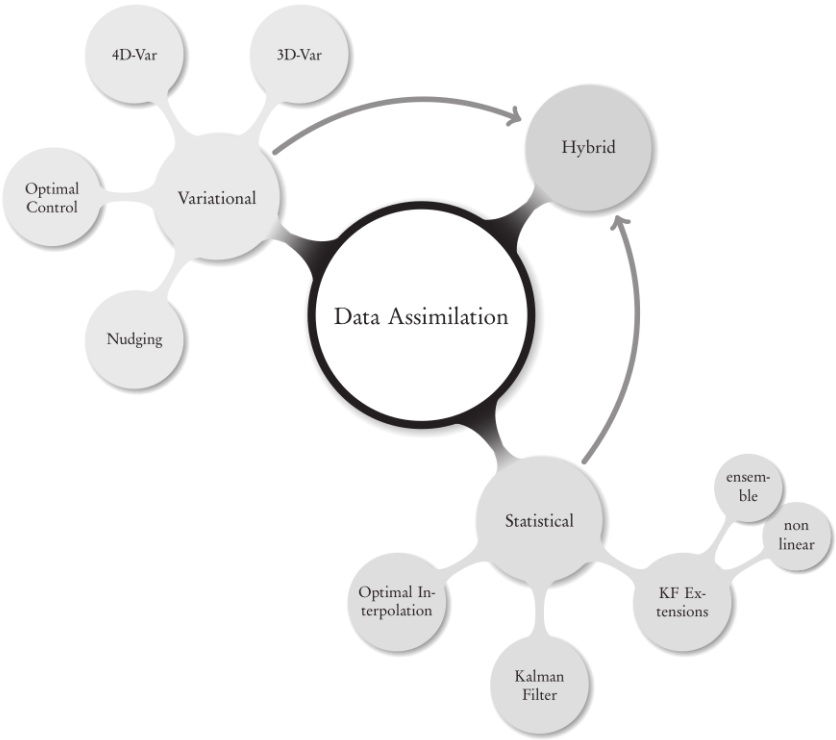


Figure 2.3: The big picture for DA methods and algorithms[3]

Connection between experiments, CFD and DA In the context of pump studies, experimental data (such as PIV or pressure measurements) provide high-fidelity but local information about the flow. CFD simulations, on the other hand, give a complete representation of the flow field but suffer from model uncertainties and parameter tuning. Data Assimilation creates the bridge between the two: it combines the predictive capability of CFD with the accuracy of experimental measurements, yielding a corrected flow representation that is both physically consistent and closer to reality. This makes DA a powerful tool for calibrating turbulence models, improving predictions, and enabling better understanding of complex flow mechanisms.

EnKF The EnKF, is a statistical Monte Carlo DA method that uses a set of model states (reffered to as ensemble member) to estimate the state of the system. The EnKF is particularly a well-suited tool for assimilating data on nonlinear and high-dimensional systems and has been successfully applied in various fields, including CFD simulations, allowing for the correction of model predictions based on observed data. It mainly consists of two steps:

- **Forecast**: time advancement of the different model states or ensemble members;
- **analysis**: correction of the model prediction by some measured observation describing the same physical phenomenon.

The maths behind the EnKF method are detailed in the subsection 2.3.3.

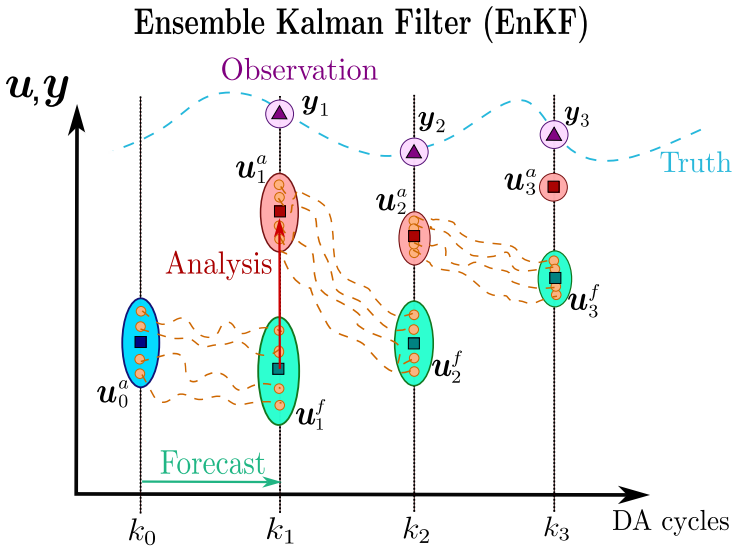


Figure 2.4: Illustration of the EnKF process (100% confidence in observations)

EnKF for DA in CFD has been used at "Laboratoire de Mécanique des Fluides de Lille" (LMFL) for a few years now.

- In 2023, Lucas Villanueva, with Miguel M.Valero and Marcello Meldi from LMFL wrote an article[16] about the amelioration of the flow around a building preforming EnKF on five CFD parameters, showing a global improvement for both the velocity field and the pressure field;
- In 2024, Tom Moussie, with Paolo Errante and Marcello Meldi from LMFL wrote an article[8] successfully optimizing the boudary condition of the academical rectangular 5:1 cylinder benchmarck using EnKF, leading to a significant improvement of the accuracy in the prediction of the turbulent flow;
- This year, Miguel M.Valero and Marcello Meldi published an article combining EnKF, Machine Learning (ML) and IBM on a turbulent channel flow [14]. It shows that this combination can gives very good and low cost results;
- During my internship, Tom Moussie submitted an article on the determination of the incidence angle of a 2D wing using EnKF, and Miguel also submitted also an article about the optimization of forcing parameters of an IBM around an 2D oscillating cylinder, showing very good results.

This work focuses on applying the EnKF method to optimize the parameters of the IBM forcing term in a centrifugal pump configuration. Compared to the previous test cases, the pump presents additional challenges: it combines turbulent boundary layer effects, rotor-stator interactions, and a complex geometry that requires around 11 million cells. These features make the flow highly unsteady and difficult to capture with traditional CFD approaches.

The main objective is to employ a DA algorithm to enhance an IBM for the simulation of a centrifugal pump. This is achieved by combining CFD simulations with experimental measurements, namely PIV data in the diffuser and pressure tap data at the impeller inlet and diffuser. The aim is to improve the predictive capability of the IBM and gain deeper insight into the pump's flow behavior.

To reach this goal, several secondary objectives are defined:

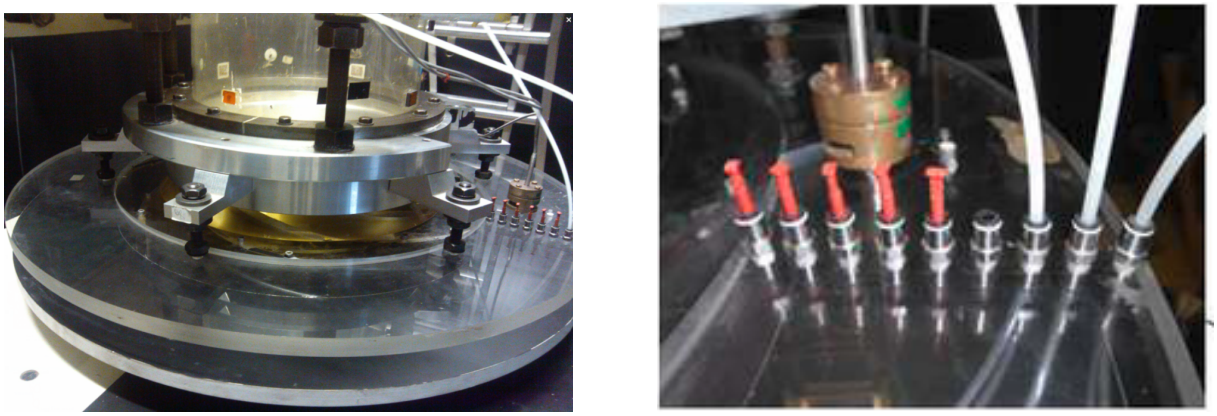
- **Assess the limitations of the baseline IBM**: investigate how well the current implementation predicts centrifugal pump flows, particularly in the diffuser and near rotating components.
- **Prepare high-fidelity experimental data**: identify, process, and validate PIV and pressure measurements to be used as reference inputs for DA.
- **Integrate an adapted DA methodology**: implement an Ensemble Kalman Filter (EnKF) within the CFD solver and ensure its compatibility with the pump test case.
- **Optimize IBM parameters via DA**: calibrate the IBM model using experimental data to improve both local flow predictions in the diffuser and global pump performance indicators.

The thesis is organized as follows:

1. A state of the art on centrifugal pumps, CFD methods (including IBM and its limitations), and DA techniques (including EnKF) is presented in section 2.1;
2. Experimental data are processed for comparison with numerical results in section 2.2;
3. The numerical framework, including governing equations, OpenFOAM, EnKF, CONES, and pump modeling, is introduced in subsection 2.3.1;
4. OpenFOAM is modified to perform first IBM-based simulations (section 3.2);
5. CONES is adapted to the pump case and DA is applied to optimize the IBM parameters (section 3.3);
6. Results of the DA are analyzed in terms of diffuser flow field and pump performance predictions (section 3.4).

2.2 Pre-processing of experimental data

As discussed in section 2.1, the RESEDA benchmark facility is located at LMFL, where PIV and pressure data have been acquired.



(a) Vaneless diffuser of the centrifugal pump

(b) Pressure taps on the diffuser

Figure 2.5: Diffuser and pressure taps of the centrifugal pump

The operating conditions of the machine were:

	Impeller characteristics	
R_1	Tip inlet radius	$141.1\,mm$
R_2	Outlet radius	$257.5\,mm$
Z	Number of blades	7
Q_d	Design flow rate	$0.236\,m^3/s$
K	Mean blade thickness	$9\,mm$
Ω	Rotating velocity	$1\,200\,rpm$
β_2	Outside blade angle (from peripheral velocity $U = R_2\Omega$)	22.5°
	Vaneless diffuser characteristics	
b_3	Diffuser width	$38.5\,mm$
R_3	Inlet radius (without leakage, i.e., $R_3 = R_2$)	$257.5\,mm$
R_4	Diffuser outlet radius	$385.5\,mm$
t	Thickness of the upper and lower sections of the diffuser	$20\,mm$

Table 2.1: Summary of the main geometrical and operating parameters of the impeller and diffuser at design conditions

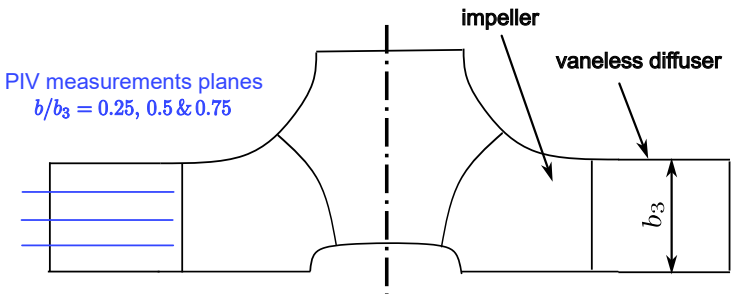


Figure 2.6: Position of the PIV planes on the diffuser

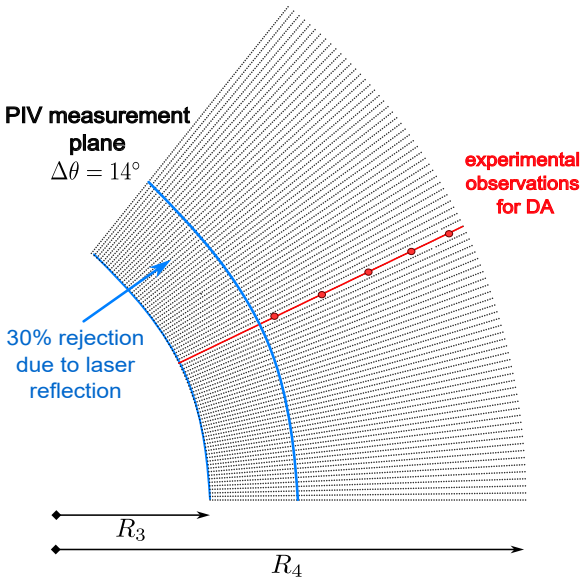


Figure 2.7: Position of the PIV points on a polar slice of the diffuser

PIV data Data is stored in a directory structure that contains six directories. The six directories correspond to two tests realised at 25%, 50% et 75% of the diffuser height. Each test was repeated twice for the same operating condition and the same height (R_1, R_2). In each directory, there are 1581 files corresponding to instantaneous ASCII data file. The ten columns are corrresponding to

$$x(m), y(m), V_x(m.s^{-1}), V_y(m.s^{-1}), r(m), \theta(rad), Vr(m.s^{-1}), V_\theta(m.s^{-1}), V_z(m.s^{-1}), \textit{Unknown}$$

And the 10,125 lines correspond to the 81 locations over x and 125 locations over y of the PIV. The PIV data is acquired in a rectangular zone of 0.08 m x 0.124 m. The cathesian coordinates are given in a local system, where the absolute origin is at the center of the pump.

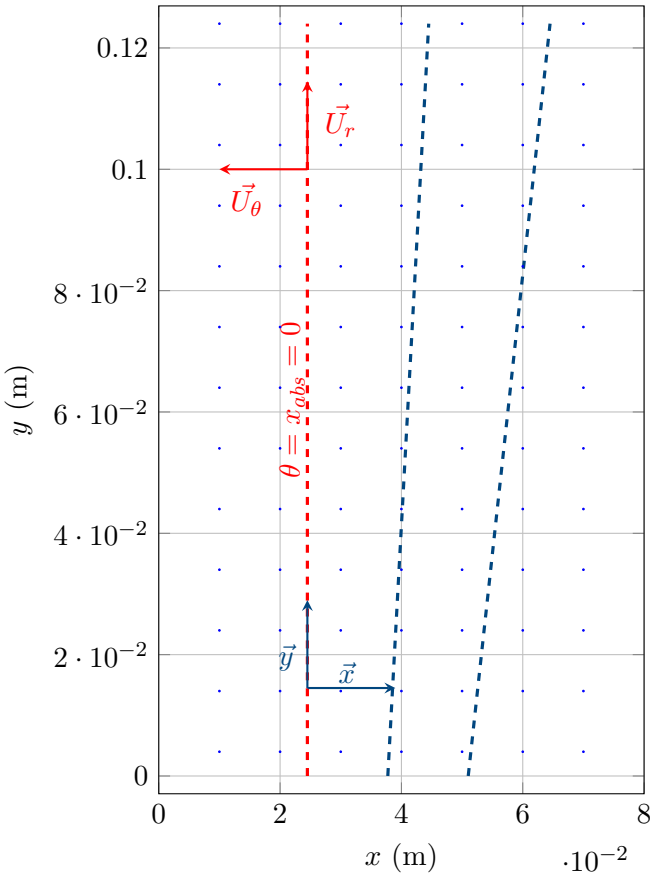


Figure 2.8: PIV acquisition zone in the RESEDA experiment, with orientation of the (r, θ) velocity components.

DA details

In this work, we aim to improve the numerical model of a centrifugal pump using experimental data obtained from PIV. On one side, we have a numerical simulation of the pump (the digital twin), and on the other side, experimental PIV measurements. To combine the strengths of both, we use data assimilation, a method that updates a numerical model using real-world data. We specifically use the EnKF, a data assimilation technique that relies on an ensemble of simulations to estimate and reduce uncertainty in the model. The process can be summarized in two main steps:

1. **Forecast Step** We run a steady-state numerical simulation of the pump using the Multiple Reference Frame (MRF) and the IBM methods. This yields an ensemble of model states, representing the expected flow field within the pump.
2. **Analysis Step** We compare the simulation results with the experimental PIV data. To do this, we interpolate the PIV measurements onto the mesh of the numerical model, creating an ensemble of observation vectors. We then compute the difference between the simulated and measured data, known as the innovation (see mathematical formulation in subsection 2.3.1). This innovation is used to update the ensemble through the EnKF algorithm, improving the numerical model.

The measurements are done as follow:

1. **Nature of the Numerical Simulation** The simulation is stationary (time-independent) and uses the MRF approach to model rotation. While the velocity field remains constant

over time, it varies in space, especially in the diffuser region. Because of the pump's geometry, we expect a periodic velocity pattern that repeats every 1/7th of a revolution (due to the 7 blades). However, this repeating pattern is not clearly visible in unsteady simulations due to wake deformation.

2. **Nature of the Experimental Data** The PIV system captures 49 images per full rotor revolution, meaning there are 7 images between two successive blades. In the rotating frame of reference, each image corresponds to a phase shift of approximately $2\pi/49$.

Ideally, to compare consistent datasets, we could run a time-resolved unsteady simulation and synchronize it with the PIV snapshots. However, for this initial study, we chose to work with a steady MRF simulation.

Pre-processing on the velocity data[3]

To address these issues, we opted to average the PIV data over time, which is equivalent to spatial (azimuthal) averaging in the rotating frame. On the simulation side, we therefore perform azimuthal averaging of the numerical results to make them consistent with the experimental data. This approach ensures that the data fed into the EnKF, both the forecast and the observations, are comparable and compatible in both spatial and statistical sense.

1. First, we need to determine how to convert the coordinates from the local system to the absolute system. We observe that between lines 55 and 56, the radius remains constant while the angular position θ shifts from below π to values above π. For the 75% plane, a similar approach is used, but between lines 65 and 66, due to experimental constraints.

We also plotted x versus θ for three different values of θ and found an intersection point at $x = 0.0245$. Therefore, we now know that for $x = 0.0245$, $\theta = \frac{\pi}{2}$.

The absolute coordinate system is then defined by the following equations:

$$\begin{cases} x_{\text{rel}} = x_{\text{abs}} + 0.0245 \\ y_{\text{rel}} = x_{\text{abs}} + R_2 \end{cases} \tag{2.1}$$

with $R_2 = 257.5$ mm

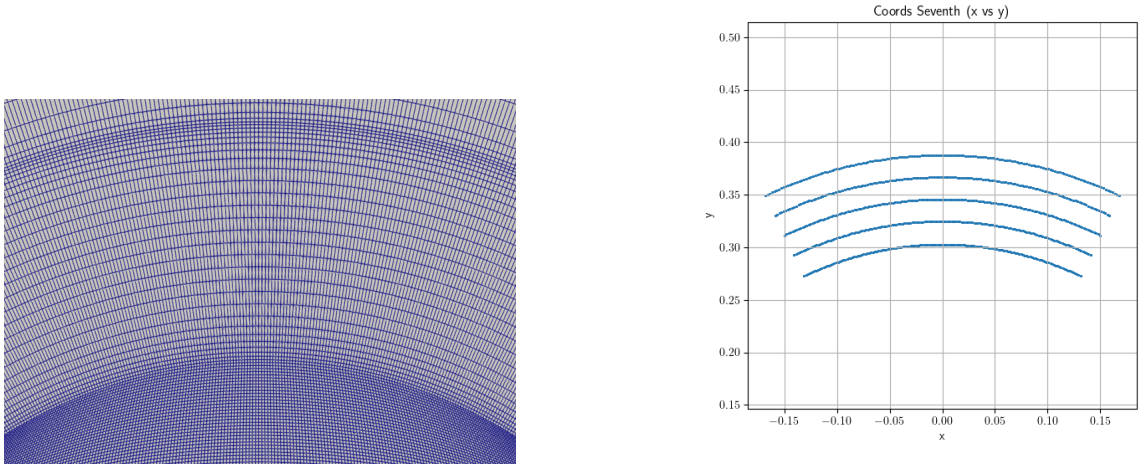
2. Average Since we are dealing with a stationary simulation (time-invariant), we perform a time average of the data. There are approximately 49 velocity maps (i.e., 49 files) per impeller rotation. To compute the time-averaged field, we average over the azimuthal angle θ.These files are distributed across six directories, each corresponding to a different acquisition plane.

3. Sampling We further average the data between radii R_1 and R_2, resulting in three files of size 81×125 lines and 10 columns. At this stage, we retain only the data along the line $x = 0.0245$. More precisely, we average the data between $x = 0.024$ and $x = 0.025$. Finally, the data are truncated at line 89, where the radius exceeds 0.3 m. This truncation is necessary due to laser reflections on the pump casing during the PIV experiments.

4.Shaping for CONES CONES keeps data in a particular way. We need to convert the data in a format that CONES can read. From 3 files (of $10 \times 81 \times 125$ elements) to the files `fieldCylindrical.txt` (of size $n_{\text{obs}} \times n_{\text{dim}} \times n_{\text{vars}}$ lines) and `obs_coordinates.txt` (of size n_{obs} lines and n_{dim} columns).

[3]In the numerical simulation, the z-axis is toward the center of the Earth, unlike in the experiment, where it is upwards. Then, the wheel rotates counterclockwise in the numerical simulation frame and counterclockwise in the experiment. Here, the numerical simulation formalism was the reference for the frame.

5.Creation of a file for the interpolation Finally, we need to create the `coords_seventh.txt` file for the interpolation function in CONES. We don't just use the coordinates of the points in the PIV data, beacause we want to average over the angle θ on the pump. The numerical simulation is $2\pi/7$ periodic, so we interpolate in 150 points over theta between 0 and $2\pi/7$, as illustrated in the Figure 2.9a. To do that, we just take the `obs_coordinates.txt` file and duplicate 150 times the points over the angle θ (between $-\pi/7$ and $\pi/7$). In our case, we chose 150 that will aproximately correspond to the shap of a cell at this radius. To be consistent with axis further, we take points between $-\pi/7 + \pi/2$ and $\pi/7 + \pi/2$. [4]



(a) Diffuser mesh

(b) Interpolation points over $2\pi/7$ degrees

Figure 2.9: Diffuser mesh and interpolation points over $2\pi/7$ degrees

Pressure data In addition to the nine pressure tapes on the diffuser, Dazin et al. [4] have also captured (Figure 2.10) the inlet and outlet pressure that allows to compute the global performance of the pump, that matches with M. Fan simulation [5]

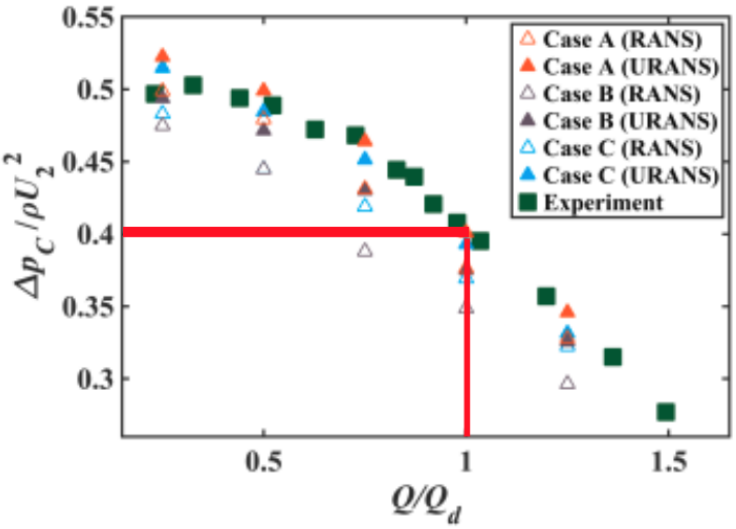


Figure 2.10: Overall pump performance versus reduced pump flow rate Q[4]

[4]Concretely, to average the data over the angle θ, we modify the interpolation function in CONES. Instead of read coords, the interpolation function read the data in a file named `coords_seventh.txt`. Then, it computes the interpolation and average the data over the angle θ (between -$\pi/7$ and $\pi/7$), by doing a sum over θ and dividing by the number of points (150). To go back from 150x5x3x3 (to detail) to 5x3x3 data. All of those computations are made in a cylindrical referencial.

As we are at $Q/Q_d = 1$, we can retreive:

$$\frac{\Delta P_P}{\rho U_2^2} = \eta \Leftrightarrow \Delta P_P = \rho U_2^2 \eta \tag{2.2}$$

For the high-fidelity value, we can find experimental data from the same pump [5], that's matches with M. Fan simulation (equal to approximately $\eta = 0.4$). So we can also compute:

$$U_2 = (\Omega \times R2)^2 = (2 \times \pi \times \frac{1200}{60} \times 0.2566)^2 = 32,2m/s \tag{2.3}$$

With U_2 the impeller tip speed computed from the impeller diameter and the rotation speed of the pump.

Finally we can compute the pressure jump at the pump level, that will be used as a reference for the data assimilation:

$$\Delta P_p = 1.225 \times 32.24^2 \times 0.4 \approx 415 \text{ Pa} \tag{2.4}$$

2.3 Numerical ingredients

2.3.1 CFD governing equations

Navier-Stokes equations CFD is based on the fundamental equations of fluid mechanics. When the fluid is assumed to be Newtonian, these equations reduce to the nonlinear Navier-Stokes partial differential equations.

These equations express three physical conservation principles, given here in their Eulerian form (i.e., fixed spatial reference frame), which is the standard in CFD because it describes how fluid properties evolve at fixed locations in space, which is ideal for mesh-based numerical simulations.

- **Mass conservation (Continuity equation):**

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{U}) = 0 \tag{2.5}$$

  with ρ the density and $\mathbf{U}$ the velocity.

- **Momentum conservation (second Newton Law):**

$$\rho\Big(\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U}\cdot\nabla)\mathbf{U}\Big) = -\nabla P + \nabla\cdot\boldsymbol{\tau} + \mathbf{f} \tag{2.6}$$

  where P is the pressure, $\boldsymbol{\tau}$ is the viscous stress tensor and $\mathbf{f}$ includes body forces (e.g., gravity, source terms).

- **Energy conservation (First law of thermodynamics):**

$$\frac{\partial(\rho E)}{\partial t} + \nabla\cdot(\mathbf{U}(\rho E + P)) = \nabla\cdot(k\nabla T) + \Phi \tag{2.7}$$

  where k is the thermal conductivity, T the temperature, and Φ the viscous dissipation and e the potential energy. $E = e + \frac{1}{2}\boldsymbol{U}^2$ is the total energy per unit mass, with:

  - e: internal energy per unit mass (also called specific internal energy)
  - $\frac{1}{2}\boldsymbol{U}^2$: kinetic energy per unit mass (specific kinetic energy)

  For an ideal gas, the internal energy is generally given by $e = c_v T$, where:

  - c_v is the specific heat capacity at constant volume
  - T is the temperature

Two main indicators of the main behaviour of the flow are the Mach number (Ma) and the Reynolds number (Re).

The Ma, is defined by the velocity of the flow divided by the speed of the sound (relative to the flow). In CFD, flow is usually considered as incompressible for Mach lower than 0.3. In our case, in atmospheric conditions,

$$Ma = \frac{R_2\Omega}{c} < 0.095 \quad \textbf{(incompressible)} \tag{2.8}$$

With $R_2 = 0.2575m$ the external radius of the impeller, $\Omega = 2\pi \times 1200/60 = 125.7rad/s$ the angular velocity of the impeller and $c = 343m/s$ the speed of sound in air at 15řC.

As we consider that we are in **incompressible** flow conditions (because we are at low Ma), Equation 2.5 becomes

$$\nabla \cdot \mathbf{U} = 0 \tag{2.9}$$

And the pressure and velocity are solved iteratively, contrary to compressible flows.

The dimensionless Re measures the ratio between inertial and viscous forces. It indicates if the flow is turbulent, laminar or both, it caratcterizes the flow regime. Let's compute the Re around where the velocity should be the highest.

$$Re = \frac{UL}{\nu} \tag{2.10}$$

U is the velocity of the fluid, L is the characteristic length and ν the kinematic viscosity of the fluid.

$$Re_{r_{tip}} = \frac{R_2^2 \Omega}{\nu} = 5.53 \times 10^5 \tag{2.11}$$

We are therefore in a **turbulent** flow regime.

Turbulence modeling The Navier-Stokes equations are exact, but in turbulent flows they cannot be solved directly due to the wide range of spatial and temporal scales. The smallest structures, the Kolmogorov scales, are where viscous dissipation occurs. Resolving them all through **DNS** is computationally unaffordable for most practical cases.

To make the problem tractable, turbulence must be modeled. Different approaches exist (Figure 2.11a[5]):

- **RANS**: the velocity field is decomposed into mean and fluctuating components. Averaging introduces the Reynolds stress tensor, which requires closure models (e.g. k-ε, k-ω, Reynolds stress models). RANS is computationally cheap and widely used for steady-state simulations, but less accurate.
- **LES**: large, energy-containing structures are resolved, while small-scale turbulence is modeled with subgrid-scale models (e.g. Smagorinsky, WALE). LES is more accurate but significantly more expensive, especially near walls (Figure 2.11c[6]).
- **DNS**: no modeling is required, since all scales are resolved, but its prohibitive cost limits it to academic studies at low Reynolds numbers.

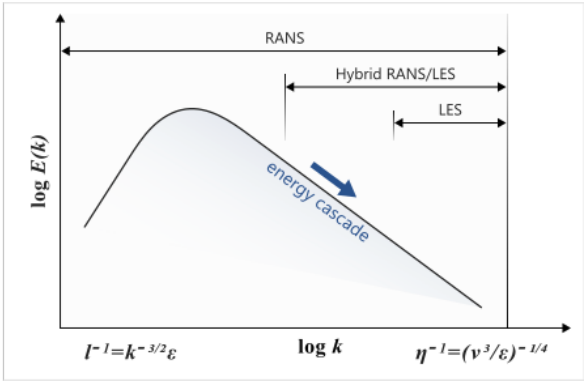
In all cases, the closure problem appears in the momentum equation through the stress tensor $\boldsymbol{\tau}$, which must therefore be modeled.

Note that DNS and LES do not rely on turbulence models in the traditional sense like RANS does; instead, they resolve directly or partially turbulence dynamics. Also, caution must be taken when interpreting visual comparisons of DNS, LES, and RANS fields (see Figure 2.11b)[7], as the underlying assumptions and modelling strategies differ.

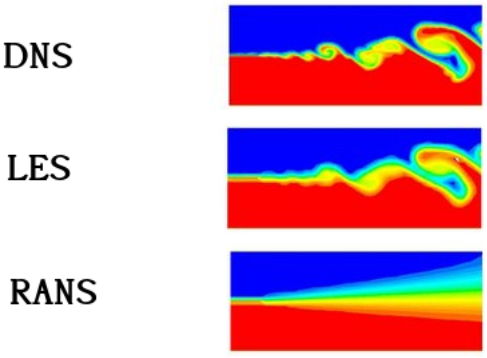
[5] https://help.altair.com/hwcfdsolvers/acusolve/topics/acusolve/training_manual/turbulence_modeling_r.htm

[6] https://www.cfdsupport.com/openfoam-training-by-cfd-support/node346/

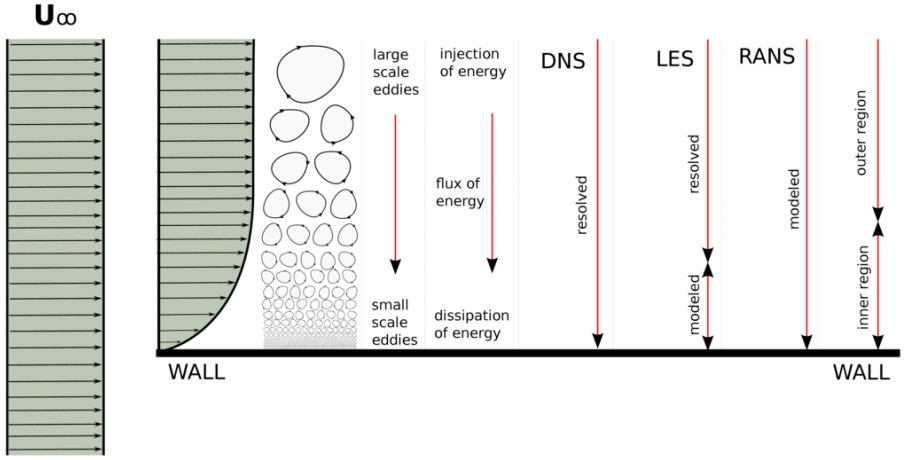
[7] https://blog.diphyx.com/comprehensive-guide-review-to-choosing-the-right-cfd-software-in-2023-features-performance-and-7ebc0623bfa6



(a) Turbulence model on the energy cascade



(b) Illustration of different CFD models



(c) Near wall modelisation of turbulence

Figure 2.11: Turbulence models

In our study, we are going to have a look on the Unsteady Reynolds-Averaged Navier-Stokes (URANS) method using the same principle as the RANS method but it's unsteady, allowing to make turn physically the pump in our case.

More precisely, in our RANS and URANS simulation, we are going to use a $k-\omega$ Shear Stress Transport (SST) turbulence model. It is a two-equation eddy-viscosity model: it combines the $K-\epsilon$ and the $K-\omega$ turbulence approximations plus a blending region between both.

$K-\omega$ is known to be a good model in the near-wall region but a bad prediction in bulk flow. $K-\epsilon$ makes a good numerical prediction in the bulk flow field but is less accurate near solid surfaces.

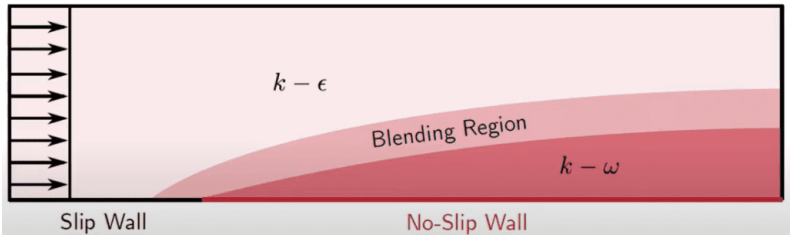


Figure 2.12: $k-\omega SST$ model

$k-\omega SST$ model is defined as follows[8][6][7]:

- Norm of the strain tensor

$$S = \sqrt{2S_{ij}S_{ij}}, \quad \text{avec } S_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) \tag{2.12}$$

- Kinematic Eddy Viscosity:

$$\nu_T = \frac{a_1 k}{\max(a_1\omega, SF_2)} \tag{2.13}$$

- Turbulent kinetic energy production term

$$P_k = \tau_{ij}\frac{\partial U_i}{\partial x_j} \tag{2.14}$$

- Turbulence Kinetic Energy:

$$\frac{\partial k}{\partial t} + U_j\frac{\partial k}{\partial x_j} = P_k - \beta^* k\omega + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_k\nu_T)\frac{\partial k}{\partial x_j}\right] \tag{2.15}$$

- Specific Dissipation Rate:

$$\frac{\partial \omega}{\partial t} + U_j\frac{\partial \omega}{\partial x_j} = \alpha S^2 - \beta\omega^2 + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_\omega\nu_T)\frac{\partial \omega}{\partial x_j}\right] + 2(1-F_1)\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_i}\frac{\partial \omega}{\partial x_i} \tag{2.16}$$

- Closure Coefficients and Auxiliary Relations:

$$F_2 = \tanh\left[\left[\max\left(\frac{2\sqrt{k}}{\beta^*\omega y}, \frac{500\nu}{y^2\omega}\right)\right]^2\right] \tag{2.17}$$

$$P_k = \min\left(\tau_{ij}\frac{\partial U_i}{\partial x_j}, 10\beta^* k\omega\right) \tag{2.18}$$

F_1 controls the blending between the $k-\epsilon$ and $k-\omega$ models. F_1 corresponds to the k_ϵ model and $F_1 = 1$ corresponds to the k_ω model.

$$F_1 = \tanh\left\{\left\{\min\left[\max\left(\frac{\sqrt{k}}{\beta^*\omega y}, \frac{500\nu}{y^2\omega}\right), \frac{4\sigma_{\omega 2}k}{CD_{k\omega}y^2}\right]\right\}^4\right\} \tag{2.19}$$

$$CD_{k\omega} = \max\left(2\rho\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_i}\frac{\partial \omega}{\partial x_i}, 10^{-10}\right) \tag{2.20}$$

$\phi = \phi_1 F_1 + \phi_2(1-F_1)$

$\alpha_1 = \frac{5}{9}, \quad \alpha_2 = 0.44$

$\beta_1 = \frac{3}{40}, \quad \beta_2 = 0.0828$

$\beta^* = \frac{9}{100}$

$\sigma_{k1} = 0.85, \sigma_{k2} = 1$

$\sigma_{\omega 1} = 0.5, \sigma_{\omega 2} = 0.856$

[8] https://www.cfd-online.com/Wiki/SST_k-omega_model

Integration into the NS equations In practice, the turbulence model modifies the momentum equation through the stress tensor $\boldsymbol{\tau}$. In RANS or URANS, $\boldsymbol{\tau}$ is decomposed into a molecular and a modeled part using the Boussinesq hypothesis:

$$\tau_{ij} = 2\mu S_{ij} - \frac{2}{3}\rho k \delta_{ij} \quad \Rightarrow \quad \tau_{ij}^{\text{RANS}} = 2\mu_{\text{eff}} S_{ij} - \frac{2}{3}\rho k \delta_{ij} \tag{2.21}$$

where $\mu_{\text{eff}} = \mu + \mu_t$ is the effective viscosity and $\mu_t = \rho \nu_T$ is the turbulent (eddy) viscosity provided by the turbulence model.

Thus, the discretized momentum equation solved in OpenFOAM is written as:

$$\rho \left(\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla)\mathbf{U} \right) = -\nabla P + \nabla \cdot \left[(\mu + \mu_t) \left(\nabla \mathbf{U} + \nabla \mathbf{U}^T \right) \right] + \mathbf{f} \tag{2.22}$$

The additional transport equations for k and ω are solved simultaneously with this momentum equation, and at each time step the updated μ_t field is used to close the system.

In other words, the turbulence model does not change the structure of the NS equations, but it adds new PDEs for k and ω and supplies the turbulent viscosity μ_t that augments the diffusive term in the discretized momentum balance.

This equation is solved in OpenFOAM (see appendix 1).

Near-wall modeling On the ω region, we can define a near-wall model where the mesh is too coarsed to capture the inner sublayer (see Figure 2.13)[9].

We can define[10] the nomalized wall ditance as:

$$y^+ = \frac{\rho u_{tau} y}{\mu} \tag{2.23}$$

where y is the distance from the wall and μ the fluid dynamic viscosity. This will allows us to know where to use a near-wall model.

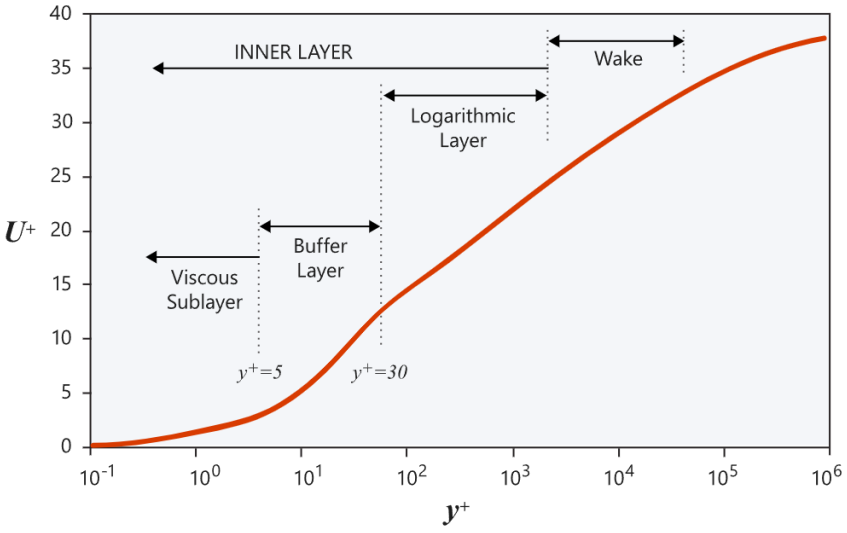


Figure 2.13: Near wall model

[9] https://2021.help.altair.com/2021/hwsolvers/acusolve/topics/acusolve/training_manual/near_wall_modeling_r.htm

[10] https://www.simscale.com/forum/t/defining-turbulent-boundary-conditions/80895, https://www.openfoam.com/documentation/guides/latest/doc/guide-bcs-wall-turbulence-omegaWallFunction.html

MRF method To simulate the rotation of the impeller in a steady-state RANS simulation, we use the Multiple Reference Frame (MRF) method. The MRF strategy consists in solving the Navier-Stokes equations in a rotating reference frame in the rotating region, while keeping the rest of the domain in a stationary (inertial) frame. The mesh and geometry remain static; only the reference frame changes.

The governing equations in the rotating frame (relative velocity) are derived from the Navier-Stokes equations in the inertial frame by applying a transformation that accounts for the rotating frame accelerations.

For an incompressible flow, the Navier-Stokes equations in the rotating frame with relative velocity $\vec{u}_R$ are:

$$\begin{cases} \frac{\partial \vec{u}_R}{\partial t} + \nabla \cdot (\vec{u}_R \otimes \vec{u}_R) + 2\vec{\Omega} \times \vec{u}_R + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) = -\nabla \left(\frac{p}{\rho}\right) + \nu \nabla^2 \vec{u}_R \\ \nabla \cdot \vec{u}_R = 0 \end{cases} \tag{2.24}$$

Where:

- $\vec{u}_R$ is the velocity in the rotating frame;

- $\vec{\Omega}$ is the angular velocity vector;

- $\vec{r}$ is the position vector;

- ρ is the fluid density;

- ν is the kinematic viscosity.

The additional terms $2\vec{\Omega} \times \vec{u}_R$ (Coriolis force) and $\vec{\Omega} \times (\vec{\Omega} \times \vec{r})$ (centrifugal force) appear due to the non-inertial frame transformation.

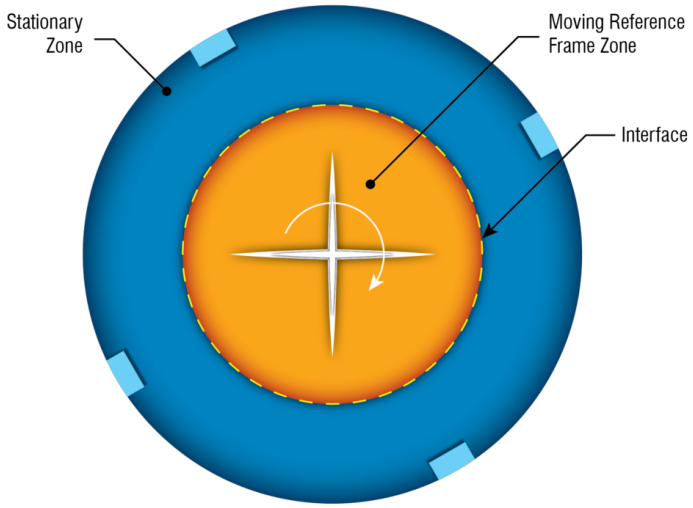


Figure 2.14: MRF strategy: the rotating region is solved in a non-inertial frame

URANS In the URANS case, the impeller is physically moving.

To do so, the CFL or the courant number has to be computed.

$$CFL = \frac{\Delta_t}{\Delta_x} \times \mathbf{U} \tag{2.25}$$

Typically set to a maximum of 1 for stability.

The CFL number can be interpreted as the fraction of the cell size that a disturbance (or wave) travels during a single time step. In this case, it corresponds to the ratio between the distance covered at the impeller tip in one time step and the space between two cells in the azimuthal direction. The CFL is computed locally at each mesh element.

IBM equations As seen in the section 2.1, different techniques exist to penalise the immersed part. In our case, we are going to use a classical penalisation method proposed by Angot et al.[2] which enters the family of continuous forcing approach. Instead of using body-fitted mesh, we are going to consider porous medium and add a forcing term on the Navier-Stokes equations (Equation 2.28) to simulate the effect of the immersed boundary on the fluid flow. This term is added only on the penalization part $\Omega_b U \Sigma_b$ (see Figure 2.15).

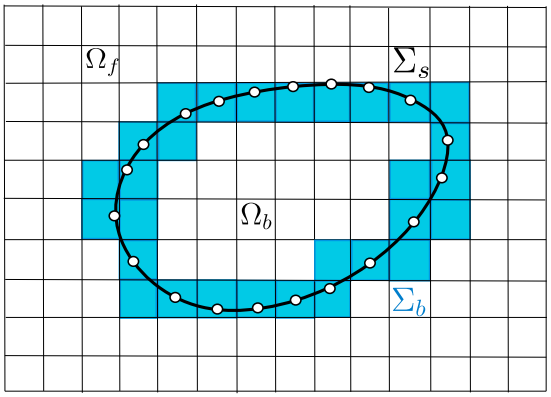


Figure 2.15: IBM regions

The Darcy's law is used in the body interface region Σ_b and the solid region Ω_b. To do so, a spatial dependent tensor $\boldsymbol{D}(\boldsymbol{x})$ must be defined. The resulting penalty term is modelled as follows:

$$\boldsymbol{f}_P = \begin{cases} \mathbf{0} & \text{if } \boldsymbol{x} \in \Omega_f \\ -\nu\, \boldsymbol{D}\,(\boldsymbol{u} - \boldsymbol{u_{ib}}) & \text{if } \boldsymbol{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \tag{2.26}$$

$\boldsymbol{u_{ib}}(\boldsymbol{x}, t)$ is a target velocity representing the immersed body's physical behaviour. In our case, $\boldsymbol{u_{ib}} = \Omega \times \boldsymbol{R}$. The components D_{ij} of the tensor $\boldsymbol{D}$ are usually controlled to obtain the best compromise between the accuracy and stability of the numerical solver. To converge to the correct result, $\mathbf{D}$ should tend to the infinite. But this lead to numerical instabilities.

We can also write Equation 2.26, once inculding in Naviers-Stokes equations, as:

$$(\boldsymbol{u} \cdot \boldsymbol{\nabla})\, \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} \underbrace{-\xi \nu \boldsymbol{D}\,(\boldsymbol{u} - \boldsymbol{u}_{ib})}_{\boldsymbol{f}_P} \tag{2.27}$$

$$\xi = \begin{cases} 0 & \text{if } \boldsymbol{x} \in \Omega_f \\ 1 & \text{if } \boldsymbol{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \tag{2.28}$$

Here, we are dividing both sides of the equations by ρ for simplification which can be considered uniform as we assume incompressible conditions. So $p = \frac{P}{\rho}$, then $[p] = M \cdot L^{-1} \cdot T^{-2}$ and $[\boldsymbol{u}] = L \cdot T^{-1}$

As we use a RANS turbulence model, in addition to the velocity, we need to penalize the k and ω coefficients.

Finally, the forcing terms are defined as follows,

$$\boldsymbol{f}_{P_u} = \xi \nu \boldsymbol{D} \left(\boldsymbol{u} - \boldsymbol{u}_{\text{ib}} \right) \tag{2.29}$$
$$\boldsymbol{f}_{P_\mathcal{K}} = \xi \nu D_\eta \left(\mathcal{K} - \mathcal{K}_{\text{ib}} \right) \tag{2.30}$$
$$\boldsymbol{f}_{P_\omega} = \xi \nu D_\omega \left(\omega - \omega_{\text{ib}} \right) \tag{2.31}$$

2.3.2 The open-source code OpenFOAM

Simulations are performed using the C++ framework OpenFOAM, used to discretize the equations previously presented. OpenFOAM is an finite-volume, open-source code which provides many tools, referred to as executables.

OpenFOAM provides a wide range of tools for solving the Navier-Stokes equations, including solvers, meshing tools, and post-processing tools. It is widely used in the CFD community and has a large user base. It has been developped since 2004 and is today composed of more than a million lines. It includes three main parts:

- **Pre-processing**, including meshing and decomposing tools;
- **Solving**, including in-built and custom solvers. Each solver is built to answer a specific CFD problem;
- **Post-processing**, including many tools and libraries to compute post-processing operations. OpenFOAM fields can be visualized with `ParaView`, an open-source visualization application.

OpenFOAM is not provided with a graphical interface, which is why a simulation is built by writing "dictionaries", in which is stored the necessary information for the simulation (mesh, turbulence model, maximum number of characteristic times, ...). See tree structure appendix 2.

The `system` directory contains the main parameters regarding the simulation progress:

- The `controlDict` dictionary defines parameters such as start and end time, time steps, data output formatting, or data average computation;
- the `blockMeshDict` dictionary defines the computational domain, mesh and boundary implementation conditions;
- the `decomposeParDict` dictionary defines the geometry and mesh decomposition, with the objective to run the simulation in parallel, using multiple cores;
- the `fvSchemes` dictionary defines the discretisation schemes used in the Central Processing Unit (CPU)s equations;
- the `fvSolution` dictionary defines the equation solvers, tolerances and other algorithm controls.

The `constant` directory contains the different physical properties of the simulation. Moreover, the subfolder `polyMesh` contains the mesh data, obtained after the `blockMesh` command line is run:

- The `momentumTransport` dictionary defines the simulation type (DNS, RANS or LES) and, in the case of RANS or LES, specifies the corresponding turbulence closure or subgrid-scale model.

– the `transportProperties` dictionary defines the transport model (e.g., Newtonian) and some transport coefficients, such as the kinematic viscosity ν;

– the `MRFProperties` dictionary defined in our case to specify the rotating zone with the origin, rotation axis and rotation speed.

The `0` directory contains the initial solution's fields and the Boundary Conditions (BC) as well. A compiled solver can be called in the terminal of the case directory to run the simulation. If it's called, the `postProcessing` directory stores data from the running simulation, such as forces and probes' data. The `processor0`, ..., `processor"N-1"` directories directly result from the geometry decomposition, which is proceeded through the `decomposeParDict` dictionary. The `dummy.foam` is an empty file which allows to visualize the simulation with `ParaView`.

More precisely, for example, to implement the IBM forcage term on the speed U, we can manually modify the momentum equation of the corresponding solver as follows:

```
// Momentum predictor step

MRF.correctBoundaryVelocity(U) // Corrects U at boundaries to include MRF effects
                               // (e.g., rotating walls)

tmp<fvVectorMatrix> tUEqn
(
    fvm::div(phi, U)            // Convective term
  + MRF.DDt(U)                  // Time derivative in the rotating reference frame
  + turbulence->ddivDevSigma(U) // Turbulent diffusion term
  + cmptMultiply(Dprime, U - UIB)    // Forcing term: Dprime * (U - UIB)
 ==
    fvModels.source(U)          // Source terms from external models
                                // (e.g., body forces), here equal to 0
);
fvVectorMatrix& UEqn = tUEqn.ref();

UEqn.relax() // Applies under-relaxation for numerical stability

fvConstraints.constrain(UEqn); // Applies physical constraints
                               // (e.g., boundary conditions)
```

With

- *cmptMultiply* the element-wise matrix multiplication;
- $D' = \xi \nu D$, with D' denoting the reduced form of D. The prime symbol is used to distinguish the non-dimensional quantity from its physical counterpart;
- U the flow velocity;
- UIB is the imposed velocity at the walls.

2.3.3 EnKF

As seen in the section 2.1, we are going to use the stochastic EnKF to assimilate the high-fidelity data. But first, let's explain the Kalman Filter[11].

[11] https://cones-dev-pe431-2dbff22738a4d54c2b947e6fa9a9a77fadd187b3bd1a85c.gitlab-pages.ensam.eu/data_assimilation.html

Kalman Filter The estimation is calculated by using a set of observations $\mathbf{y}$, a prior state $\mathbf{x}$, and a linear model $\mathbf{M}$. The estimation state is obtained through two operations, the forecast and the analysis, respectively using the superscript f and a.

The system state $\mathbf{x}_k$ is as follows:

$$\begin{bmatrix} \mathbf{u}_k \\ \boldsymbol{\theta}_k \end{bmatrix} \in \mathbb{R}^{(n_{\text{vars}} \cdot n_{\text{cells}} + n_\theta) \times m} \tag{2.32}$$

where:

- $\mathbf{u}_k$ is the state matrix (can contain velocities, pressure, temperature...) of the domain at the time instant k, which we don't care about in our case;
- $\boldsymbol{\theta}_k$ are the model parameters;
- m is the number of ensemble members.

The observation data $\mathbf{y}_k$ is defined as:

$$[\mathbf{y}_k]_{n_{\text{obs}},m} \sim \mathcal{N}(y_k, \mathbf{R}_k) \tag{2.33}$$

Observation data may be built using:

- Velocity, pressure, temperature or other physical observed values;
- Some discret model parameters such instantaneous global force coefficients C_D, C_L, C_f (in our case, it will be five IBM forcing parameters).

Prediction step:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{M}_k \hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}_k \mathbf{u}_k \tag{2.34}$$
$$\mathbf{P}_{k|k-1} = \mathbf{M}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^\top + \mathbf{Q}_k \tag{2.35}$$

Update step:

$$\tilde{\mathbf{y}}_k = \mathbf{z}_k - \mathbf{H}_k \hat{\mathbf{x}}_{k|k-1} \tag{2.36}$$
$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k \tag{2.37}$$
$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1} \tag{2.38}$$
$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \tilde{\mathbf{y}}_k \tag{2.39}$$
$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \tag{2.40}$$

Where:

- $\hat{\mathbf{x}}_{k|k-1}$: predicted state at k
- $\hat{\mathbf{x}}_{k-1|k-1}$: updated state at $k-1$
- $\mathbf{M}_k$: state transition matrix
- $\mathbf{B}_k$: control input matrix
- $\mathbf{u}_k$: control input
- $\mathbf{P}_{k|k-1}$: predicted covariance
- $\mathbf{P}_{k-1|k-1}$: updated covariance at $k-1$
- $\mathbf{Q}_k$: process noise
- $\mathbf{z}_k$: measurement
- $\mathbf{H}_k$: observation matrix

- $\tilde{\mathbf{y}}_k$: innovation
- $\mathbf{S}_k$: innovation covariance
- $\mathbf{K}_k$: Kalman gain
- $\mathbf{R}_k$: noise measurement
- $\mathbf{I}$: identity matrix
- $\hat{\mathbf{x}}_{k|k}$: updated state
- $\mathbf{P}_{k|k}$: updated covariance

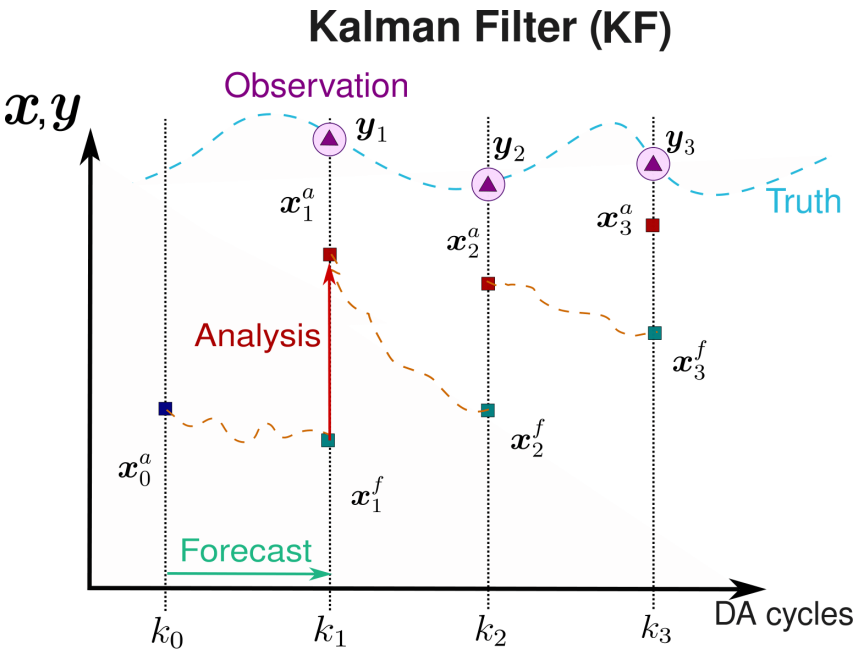


Figure 2.16: Kalman Filter process illustration

As shown in Figure 2.16, the Kalman Filter alternates between two steps: the forecast, where the model predicts the next state, and the analysis, where this forecast is corrected using available observations. The analysis step brings the model trajectory closer to the truth by optimally combining model and measurements.

Kalman filter presents two main drawbacks for CFD applications. First, it assumes linear dynamics and Gaussian errors, which is not valid when $\mathbf{M}$ represents the Navier–Stokes equations, as these are highly nonlinear. Second, it requires storing and updating the covariance matrix $\mathbf{P}_k$, whose dimension is $(n_{vars} \times n_{cells}, n_{vars} \times n_{cells})$, making it computationally prohibitive on large CFD meshes.

Ensemble Kalman Filter The Ensemble Kalman Filter overcomes the limitations of the classical Kalman Filter (KF) by using an ensemble of states to represent the uncertainty instead of explicitly propagating the full covariance matrix. Each ensemble member evolves independently according to the model equations, and the sample covariance of the ensemble is then used to update the state when new observations become available.

In practice:

- During the forecast step, all ensemble members are advanced in time by the CFD solver.
- In the analysis step, each ensemble member is corrected using the innovation (observation minus forecast), weighted by the ensemble covariance.

The EnKF is particularly well suited for high-dimensional systems, since it avoids the explicit computation and storage of $\mathbf{P}_k$. Its Monte Carlo formulation also makes it more robust to

non-linear dynamics compared to the standard KF. A known drawback is the tendency to underestimate the analysis covariance $\mathbf{P}_k^a$, but techniques such as covariance inflation and localisation (introduced later in this subsection) are commonly applied to mitigate this issue.

We saw previously in this subsection that today, we can theorically run a DNS which resolves exactly the Navier-Stokes equations, but it is still too expensive to be used in practice. Therefore, we generally use LES or RANS models to simulate the flow. The EnKF method can be applied to these models to adjust specific model parameters or flow fields, enhancing the overall physical consistency of the simulation.

The following matrices are needed to build the EnKF algorithm:

1. **Observation covariance matrix $\mathbf{R}_k$**: we assume Gaussian, non-correlated uncertainty for the observations ($\sigma_{i,k}$ is the standard deviation for the variable i at time instant k).

$$[\mathbf{R}_k]_{n_{\text{obs}},n_{\text{obs}}} = \sigma_{i,k}^2 \, [\mathbb{I}]_{n_{\text{obs}},n_{\text{obs}}}$$

2. **Sampling matrix $\mathcal{H}\left(\mathbf{x}_i^f\right)$**: it is the projection of the model into the position of the observations.

$$\left[\mathcal{H}\left(\mathbf{x}_i^f\right)\right]_{n_{\text{obs}},m}$$

3. **Anomaly matrices** for the system's state $\mathbf{x}_i^f$ and sampling $\mathcal{H}\left(\mathbf{x}_i^f\right) = \mathbf{s}_i^f$ matrices. They measure the deviation of each realization with respect to the mean.

$$\mathbf{X}_i = \frac{\mathbf{x}_i - \bar{\mathbf{x}}}{\sqrt{m-1}}, \quad \mathbf{S}_i = \frac{\mathbf{s}_i - \bar{\mathbf{s}}}{\sqrt{m-1}}$$

with

$$[\mathbf{X}]_{n_{\text{vars}} \times (n_{\text{cells}}+n_\theta),m}, \quad [\mathbf{S}]_{n_{\text{obs}},m}$$

4. **Kalman Gain matrix**:

$$\mathbf{K} = \mathbf{X}\mathbf{S}^\top \left(\mathbf{S}\mathbf{S}^\top + \mathbf{R}\right)^{-1}$$

with

$$[\mathbf{K}_k]_{n_{\text{vars}} \times (n_{\text{cells}}+n_\theta),n_{\text{obs}}}$$

For RANS representation, there is no physical time advancement of the model, so each instant k in the 1 is represented through a different interation of the solver.

Observation data y can be some local sensors (such as PIV data) or global variables (such as performance)

With $\overline{\boldsymbol{x}}_k^f = [\overline{\boldsymbol{u}}_k^f, \overline{\theta}_{k-1}]^T$ and $\boldsymbol{x}_{i,k}^f = [\boldsymbol{u}_{i,k}^f, \theta_{i,k-1}]^T$

In our case, $\overline{\boldsymbol{x}}_k^f = [\overline{\theta}_{k-1}]^T$, $\boldsymbol{x}_{i,k}^f = [\theta_{i,k-1}]^T$, $\lambda_k = 0$ and $\boldsymbol{y}_k = \boldsymbol{y}$.

Some *deterministic inflation* is included to increase the variability of the state predicted by the EnKF:

$$\boldsymbol{\alpha}_{i,k}^a = \overline{\boldsymbol{\alpha}}_k^a + \lambda_k \left(\boldsymbol{\alpha}_{i,k}^a - \overline{\boldsymbol{\alpha}}_k^a\right) \tag{2.41}$$

where $\boldsymbol{\alpha}$ can be either the state vector $\boldsymbol{u}$ or the model parameters θ.

The Figure 2.17 illustrates of the deterministic inflation, that increase the variability of the member ensemble to avoid a too fast collaps and permit to have more chance to find a lower local (or in a better case, global) minimum solution:

Algorithm 1 Scheme of the EnKF used in the present analyses for parameter optimization.

Input: $\mathcal{M}$ (model); $\boldsymbol{R}$ (observation covariance); $\boldsymbol{y}_k$ (observations);
$\boldsymbol{x}_{i,0} = [\boldsymbol{u}_{i,0}, \theta_{i,0}]^T$ (initial prior states with $\boldsymbol{u}_{i,0} \sim \mathcal{N}(\overline{\boldsymbol{u}}_0, \boldsymbol{P}_0)$ and $\theta_{i,0} \sim \mathcal{N}(\overline{\theta}_0, \boldsymbol{P}_0)$)
$k = 1, 2, ..., K$ is the iteration index; $i = 1, 2, ..., N_e$ is the ensemble index
for $k = 1$ to K **do**
  for $i = 1$ to N_e **do**
    Advance in time: $\boldsymbol{u}^f_{i,k} = \mathcal{M}(\theta_{i,k-1})\, \boldsymbol{u}_{i,k-1}$
    Observation: $\boldsymbol{Y}_{i,k} = \boldsymbol{y}_k + \boldsymbol{e}_i$, $\boldsymbol{e}_i \sim \mathcal{N}(0, \boldsymbol{R})$
    Predicted observation: $s_{i,k} = \mathcal{H}(\boldsymbol{u}^f_{i,k})$
    Ensemble means: $\overline{\boldsymbol{x}}^f_k = \frac{1}{N_e}\sum_{i=1}^{N_e} \boldsymbol{x}^f_{i,k}$, $\overline{s_{i,k}} = \frac{1}{N_e}\sum_{i=1}^{N_e} s_{i,k}$
    Anomaly matrices: $\boldsymbol{X}_k = \frac{\boldsymbol{x}^f_{i,k} - \overline{\boldsymbol{x}}^f_k}{\sqrt{N_e - 1}}$, $\boldsymbol{S}_k = \frac{s_{i,k} - \overline{s_{i,k}}}{\sqrt{N_e - 1}}$
    Kalman gain: $\boldsymbol{K}_k = \boldsymbol{X}^f_k (\boldsymbol{S}_k)^T [\boldsymbol{S}_k (\boldsymbol{S}_k)^T + \boldsymbol{R}]^{-1}$
    Update: $\boldsymbol{x}^a_{i,k} = \boldsymbol{x}^f_{i,k} + \boldsymbol{K}_k (\boldsymbol{Y}_{i,k} - \mathcal{H}(\boldsymbol{u}^f_{i,k}))$
    (Deterministic) covariance inflation: $\boldsymbol{x}^a_{i,k} = \overline{\boldsymbol{x}}^a_k + \lambda_k (\boldsymbol{x}^a_{i,k} - \overline{\boldsymbol{x}}^a_k)$
  end for
end for

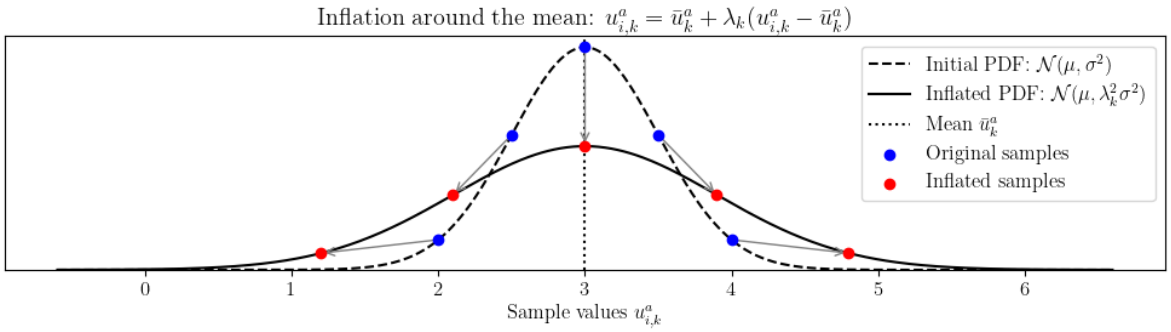
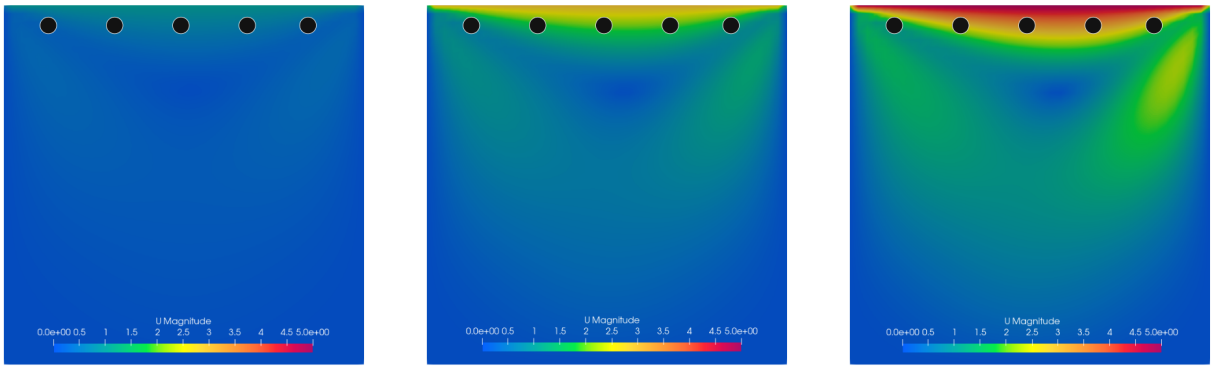


Figure 2.17: Deterministic inflation

I coded it but it finally was not needed for the centrifugal pump case.

Example A good example based on the simple and academic *cavity test case* illustrates the DA with the EnKF well. The cavity test case is a simple flow in a square cavity, where the flow is driven by a moving lid. The flow is characterized by a recirculation zone and a vortex in the center of the cavity. The EnKF method can be used to assimilate data from the flow, such as velocity or pressure measurements, to improve the accuracy of the simulation. The EnKF method can also be used to estimate the parameters of the model, such as viscosity or turbulence model coefficients, to improve the physical accuracy of the simulation.

In our test case, we are going to optimise the velocity wall condition. To do so, we use the result on 5 locations of a precise converged simulation with a to boundary condition of 5 m.s^{-1} as a reference. We will then assimilate data from the flow, at the 5 observation points (see Figure 2.18a). We start from a simulation with a wall condition of 1 m.s^{-1}, which is not the correct value. The EnKF method will then be used to correct the wall condition by assimilating the data from the flow at the 5 observation points. The EnKF method will iteratively update the wall condition until it converges to the correct value, which is 5 m.s^{-1} in this case. After only three EnKF iterations, the wall condition converges to the correct value (with a relative error of 0.05 m.s^{-1}).



(a) Initial state (b) After one EnKF it. (c) Converged after 2 EnKF it.

Figure 2.18: Evolution of the state through EnKF iterations (●: observation point)

2.3.4 The CONES library

CONES is a library coded in Python/C++, that has been developed since 2022 at LMFL. It is the only code that permits online DA in CFD codes (it uses Message Passing Interface (MPI) process to do it). It's parallelized and can be used on supercomputers using batch code as the on in appendix 4. It is mainly designed to work with OpenFOAM and DA codes as EnKF.

A specific directory structure must be observed, as outlined in Figure 2.19. Let's take our OpenFOAM/EnKF example as support to explain CONES. Concretely, on one side, m OpenFOAM simulations are running with run at the same time on n processors and on the other side, the EnKF runs on e processors (usually only one is enough). At the begining, all variables are initialized on both sides, using MPI process. Then, all simulation run at same time (with different initial states). Sampled data ($H(u_k^f)$) and $\begin{bmatrix} \mathbf{u}_k \\ \boldsymbol{\theta_k} \end{bmatrix}^f$ are send to the EnKF. Using y_k, EnKF algorithm can compute and send back the updated velocity field (not computed in our case) and the parameters θ_k^a to the different ensemble members (each are different befor converging). With these new parameters, the loop can be done again.

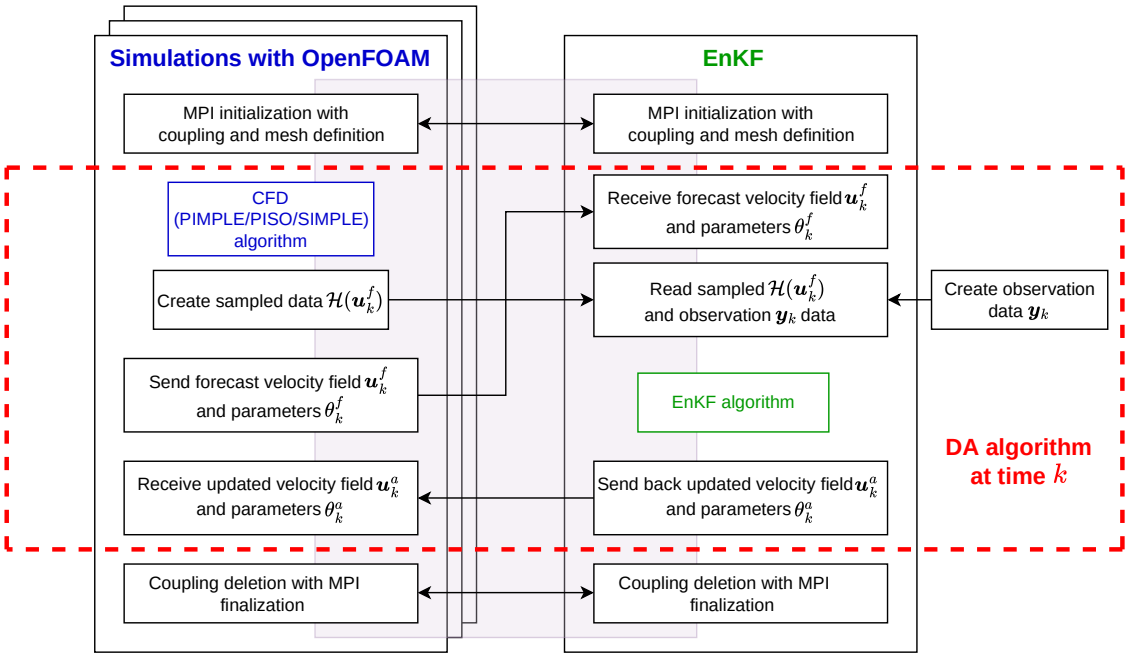


Figure 2.19: CONES structure

Chapter 3

Application

3.1 Pump modelling

Figure 3.1 illustrates the model of the pump on a yz plane, for $x = 0$. The suction pipe has a length of $H = 10R_1$, allowing the near-wall boundary lawer to develop. Then, a cylinder, where the IBM and MRF will be applied, containing the impeller, is defined (the impeller in .stl can be find on github[1]). At the entrance of the impeller/outlet of the suction pipe junction and at the outlet of the impeller/inlet of the diffuser junction, there was leakage as the whole impeller that was moving during the experiment. We assume a zero-leakage as we aren't in a very hight mass flow rate.

In our simulation, the z-axis aligns with the rotational axis, and the reference frame is centred at the impeller. The plane at $z = 0$ divides the diffuser into two identical halves.

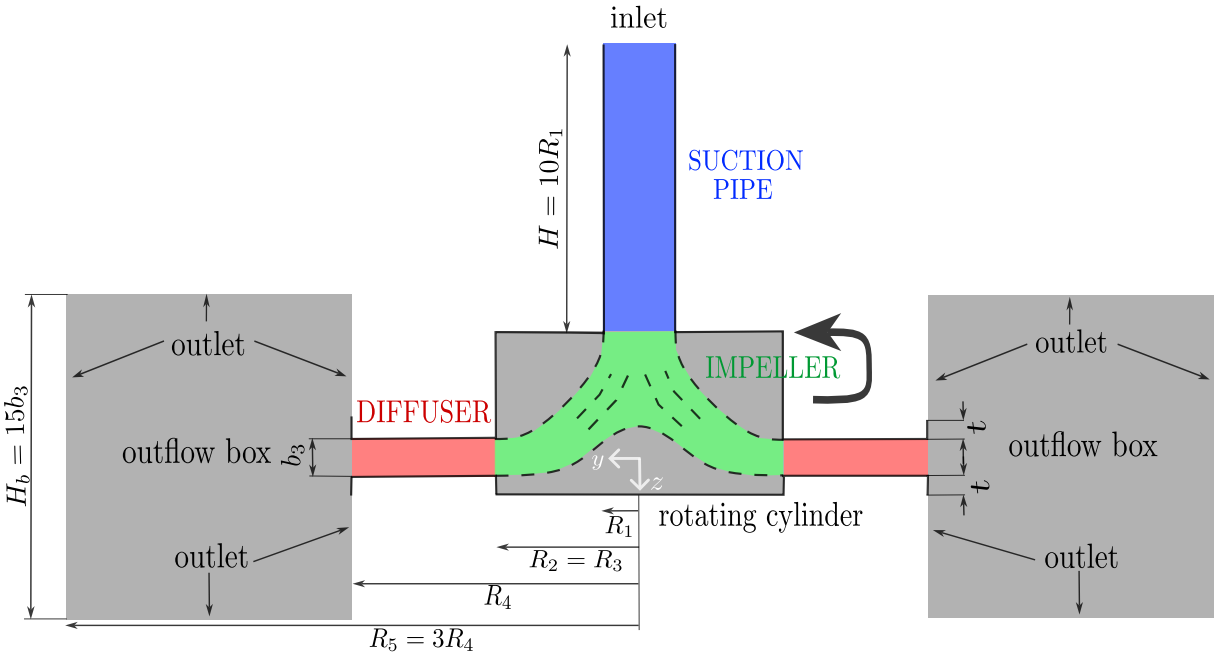


Figure 3.1: Model of the RESEDA benchmark

[1] https://github.com/fromano88/CentrifugalPump

Boundary conditions are defined as follows:

- Inlet: velocity inlet
- Outlet: pressure outlet
- Wall: no-slip wall

To process CFD, we need to discretize the numerical model of the pump. The mesh is generated using the `blockMesh` utility, which is a built-in OpenFOAM tool. It is composed of around 11 million hexahedral cells (quasi only, excepted some interface, cf Figure 3.2). We don't need complex mesh because we don't need to fit the geometry of the impeller as we use IBM.

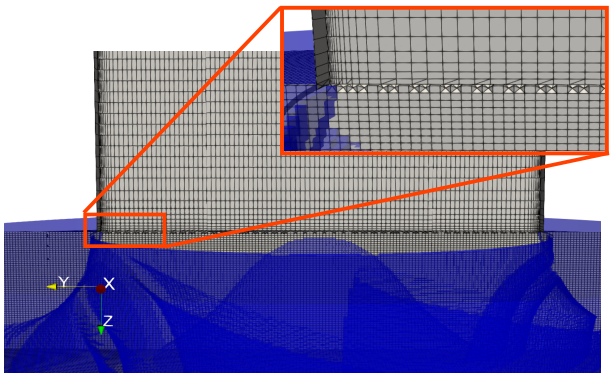


Figure 3.2: Mesh interface between the pipe and the impeller

Regarding the near-wall treatment, at the suction pipe, $y^+ \approx 1$, enabling direct resolution of the viscous sublayer and imposition of boundary conditions without wall functions. On the other hand, the highest y + value in the body-fitted regions corresponds to $y^+ \approx 83$, so some of the mesh elements (mainly close to the lower wall of the diffuser) are located in the logarithmic region, thus requiring wall functions. Within the rotating cylinder and around the impeller, the mesh resolution yields $y^+ \approx 100$. The friction velocity U_τ is assumed to be around 20 - 25 times smaller than the peripheral velocity U_2 . The blades are discretised to ensure an average of six mesh elements across their solid thickness, preventing excessively thin regions.

The penalisation coefficients used in the Equation 2.29 are defined as follows:

$$\boldsymbol{u}_{\mathrm{ib}} = \cancel{\boldsymbol{u}_r} + \boldsymbol{\Omega} \times \boldsymbol{r} \tag{3.1}$$

$$\mathcal{K}_{\mathrm{ib}} = 0 \tag{3.2}$$

$$\omega_{\mathrm{ib}} = \left(\omega_{\mathrm{vis}}^2 + \omega_{\mathrm{log}}^2\right)^{1/2} \tag{3.3}$$

where $\boldsymbol{u}_r = 0$, $\omega_{vis} = \frac{6\nu}{\beta_1 y_{\mathrm{ref}}^2}$ is the viscous sublayer and $\omega_{log} = \frac{\sqrt{\mathcal{K}}}{C_\mu \kappa y_{\mathrm{ref}}}$ is the logarithmic region (see Figure 2.13).

Knowing the mesh and the impeller velocity, based on the Equation 2.25, we can compute the Δ_t at the tip of the impeller, where the velocity is the highest and Δ_x the lowest.

- In our model, Δ_x is the distance between two cells in the azimuthal direction at the tip of the impeller, which is given as 1.5 mm or 0.0015 m.

- $\|U_\infty\|$ is the magnitude of the maximal velocity usually "at infinity distance" from the body, which in our case can be approximated by the tangential velocity at the tip of the impeller.

$$\|U_\infty\| = \Omega \cdot r_{tip} \tag{3.4}$$

where:

- ω is the angular velocity in radians per second,

- r_{tip} is the radius at the tip of the impeller.

Assuming the radius at the tip of the impeller is $r = 0.2566$ m, we can compute the tangential velocity:

$$\|U_\infty\| = 125.66 \text{ rps} \cdot 0.2566 \text{ m} = 32.24 \text{ m/s} \tag{3.5}$$

Assuming the distance between two cells in the azimuthal direction at the tip of the impeller is $\Delta_x = 0.0015$ m (which is 1.5 mm) we can now compute the time step Δ_t:

$$\Delta_t = \frac{0.5 \cdot 0.0015 \text{ m}}{32.24 \text{ m/s}} = 2.3 \times 10^{-5} \text{ s} \tag{3.6}$$

3.2 IBM preliminary results

Once the pump model is implemented in OpenFOAM, we can run the simulation with the IBM. After empirical tests and with experience of the laboratory, we found that the parameters $\mathbf{D} = (3e7, 3e7, 3e7), D_k = 3e7$ and $D_\omega = 3e7$ are a good starting point for the IBM parameters. And it verifies a converged CFD solution.

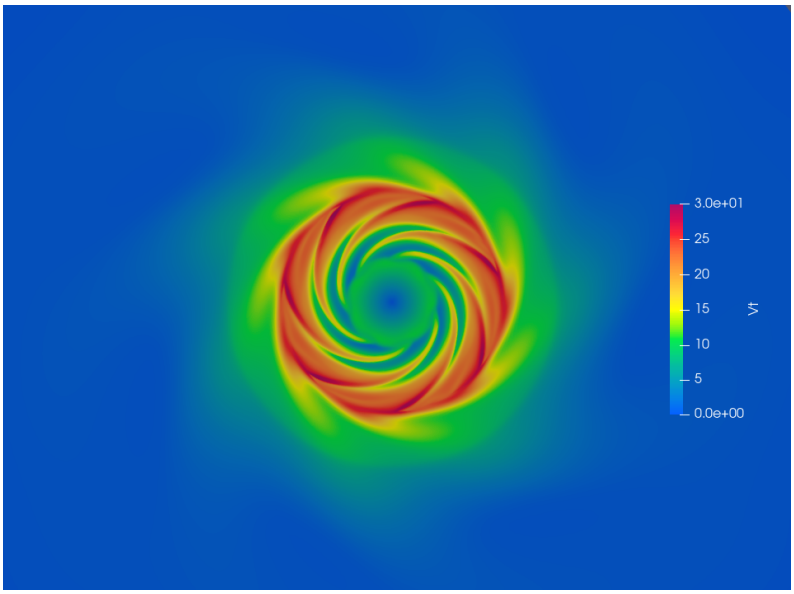


Figure 3.3: First results of the IBM for all parameters equal to 3e7. Slice at $z = 0$ for $\mathbf{D} = (3e7, 3e7, 3e7), D_k = 3e7, D_\omega = 3e7$

With the conditions specified in the Table 2.1, and a value of 3e7 for the all five forcing parameters, we obtain these velocity profils comparing the velocities of the simulation and of the PIV data. Both are averages over time and over the angle θ:

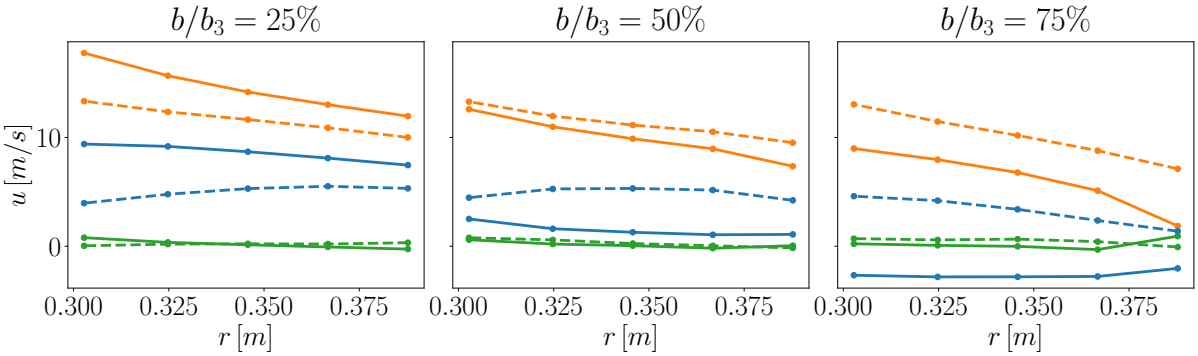


Figure 3.4: Comparison between the profiles inside the diffuser for the numerical simulation (continuous lines), and the PIV data (dashed lines). In particular, (——) represents the radial velocity u_r, (——) depicts the azimuthal velocity u_θ, and (——) illustrates the vertical velocity u_z. $\mathbf{D} = (3e7, 3e7, 3e7), D_k = 3e7, D_\omega = 3e7$

As seen in Figure 2.7, we use five observation points on the three PIV plans. Our goal is to make these curves fit as much as possible.

We have already seen that there is a notable difference beteween the experimental and numerical radial velocity in the plan at 75%. The numerical simulation predicts a negative

velocity instead of a positive one. This is explained by a non physical recirculation bubble as you can see in the Figure 3.6 deeper explained in the following part.

A good indicator of the accuracy of the simulation is the performance of the pump. It's directly linked to the pressure difference between the outlet and the inlet. As seen in the previous section 2.2, we should obtain around 415Pa. In our case, we obtain a pressure difference of around 306 Pa, far from what we expect.

3.3 First simple application of the DA

As represented on the Figure 2.19, we need to adapt the OpenFOAM and CONES codes to apply the EnKF on the centrifugal pump case. Among others, the MPI exchanges are coded.

Modifications on OpenFOAM Previously, OpenFOAM has been adapted to our case, including the implementation of the IBM and MRF methods, as well as the definition of the mesh and boundary conditions. Now, we need to adapt OpenFOAM to work with CONES, by including MPI processes that will allow to run a simulation with variable IBM parameters and that will be able to return interpolated averaged velocities to CONES.

Adaptations on CONES Although CONES is already adapted to work with OpenFOAM, we still need to adjust the code to fit our specific observations.

Test case Finally, we have to initialize a repository for the test case, i.e., the centrifugal pump. This includes defining the initial parameters, the observation points, and the initial state of the ensemble members, including a `conesDict`.

Parameters To determine the variance for the initial perturbation, two perspectives were considered. The first one is the experience at lab; it shows that an initial perturbation of around 5% of variance is good. The other one is by ourself: the target was that, around only one ensemble member differs by maximum approximately 1×10^7 from the previous time step, for parameter values of order 3×10^7. With 30 ensemble members, this corresponds to about 95% of the values (i.e., within 2σ) being contained in the interval $[3 \times 10^7 \pm 1 \times 10^7]$. This leads to a standard deviation of $\sigma = 0.5 \times 10^7$, or about 17%. We finally chose a variance of 5% for the first test.

Let us introduce the **observation window**. The observation window is defined by observing the number of iterations required for the influence of the IBM forcing to reach the diffuser and taking the form of the converged result. Turbulence parameters were not considered for defining this window, since their effect is present throughout the domain and their influence is almost immediate (although their convergence is slower).

The number of iterations between two EnKF analysis phases is now determined by monitoring how many iterations are required for a parameter perturbation (e.g., 34% change) to significantly affect the solution in the diffuser. What matters here is not full convergence, but that the solution starts to follow the correct trend. For turbulence parameters, this timescale is expected to be much shorter, since the forcing is applied everywhere in the domain. We determined a number of 800 iterations for the initial observation window.

The `conesDict` and the `controlDict` contains the parameters of the EnKF algorithm, such as the observation window and the following ones: in our case, we set the number of ensemble members to 30, no inflation, no state optimisation (only parameters), and an absolute observation noise on the velocity of 0.2 m/s (equivalent to 1.5% of the maximum observed velocity). These parameters must be adjusted depending on the specific requirements of the simulation and the characteristics of the data. The number of iterations (800) was determined by testing the convergence trend of a single OpenFOAM simulation when parameters are modified by 34% (i.e., seven times the standard deviation of the initial parameter perturbation). Since the stan-

dard deviation may temporarily increase with iterations, care must be taken when choosing this number. An extract of the `conesDict` is given in appendix 3.

First results

After letting the simulation run for around 50 iterations, we obtain the first results of the optimisation with CONES. Usually, around 200 iterations are needed to converge but as much iterations are not needed for a first result and cost a lot of CPU time (1270 CPU $\times$ hours per EnKF iteration). The parameters are optimised to fit the PIV data, and the velocity field is represented on a slice at $z = 0$. The resulting parameters are the following: $\mathbf{D} = (4e7, 2e7, 4e7), D_k = 3e7$ and $D_\omega = -3e7$. The sum of the Normalized Root Mean Square Error (NRMSE) on the velocity field decreased, of around 20%. The global performance of the pump didn't change significantly.

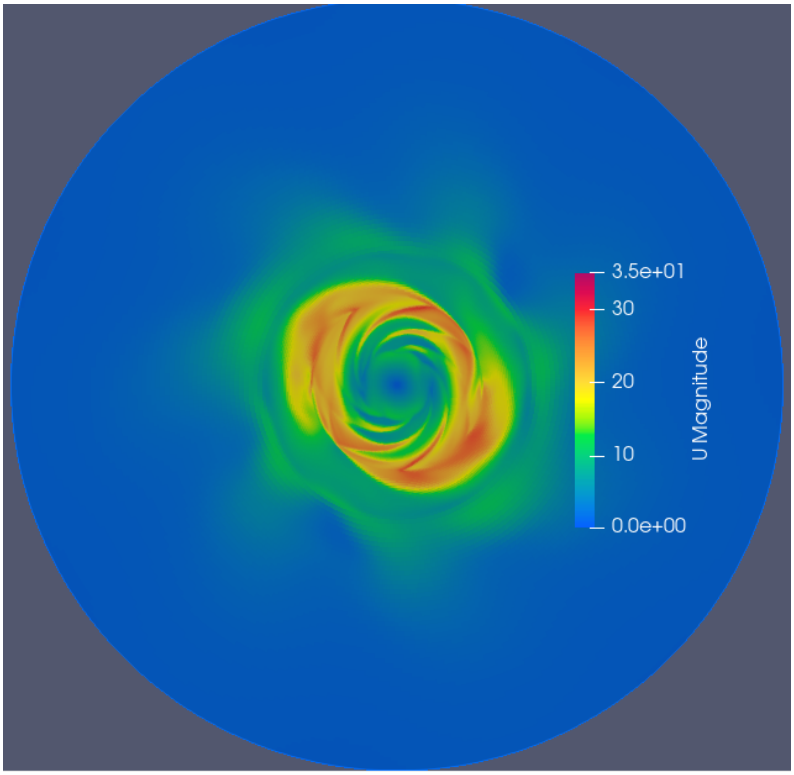


Figure 3.5: First results of CONES optimisation.
Slice at $z = 0$, with $\mathbf{D} = (4e7, 2e7, 4e7), D_k = 3e7$ and $D_\omega = -3e7$

We would expected $D_x = D_y$ because the geometry is symetrical, but we see that the optimisation converges to a solution where $D_x \neq D_y$. The local minimum found by the optimisation gives us a unsatisfactory solution. Also, D_ω is negative, which does not correspond to physical results.

3.4 Analysis and Results

Looking in more detail at the physics of the pump CFD optimised by DA, we realize that the IBM method as applied here does not give us a physical result at all and the DA does not allow us to improve this result. It let the flow pass through the external part of the impeller (see Figure 3.6), which is not physical. This imply a recirculation bubble in the diffuser that we would like to avoid.

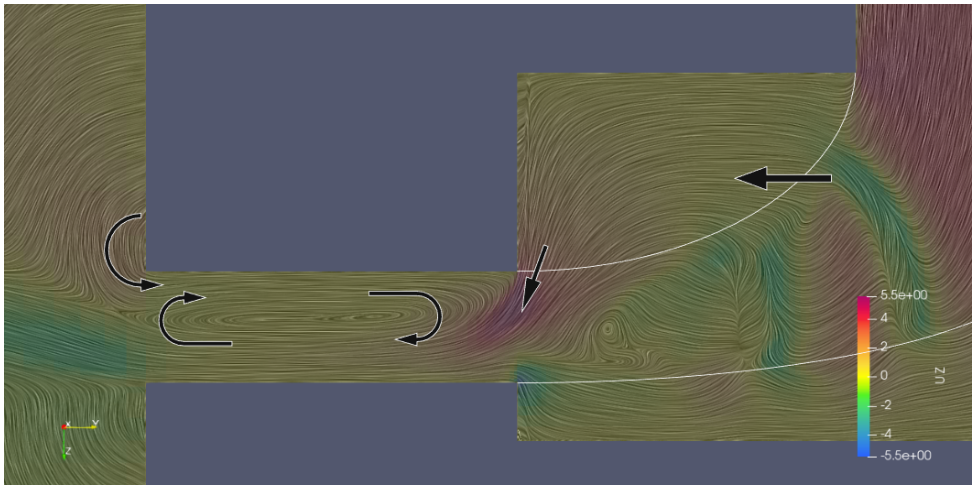


Figure 3.6: Formation of the recirculation bubble in the diffuser

To try to counter this problem, a few strategies are deployed one by one, tested separately or in combination (each DA analysis costs a lot of CPU time, so we are limited in the number of analysis we can do):

Strategie 1. $\boldsymbol{D_x = D_y}$ The strategy is to force $\boldsymbol{D_x = D_y}$ (it helps the convergence by the way, by reducing the number of parameters to optimise). In that case, the parameters vector is reduced to $\mathbf{D} = (\boldsymbol{D_x}, \boldsymbol{D_z}, \boldsymbol{D_k}, \boldsymbol{D_\omega})$.

This strategy works well, until parameters goes up to make the simulation diverge (see Figure 3.7). So it does not solve the problem of the recirculation bubble, but it helps to have a more physical result. The sum of the NRMSE on the velocity field decreased, of around 15% before collapsing again to a result with no circular repetition.

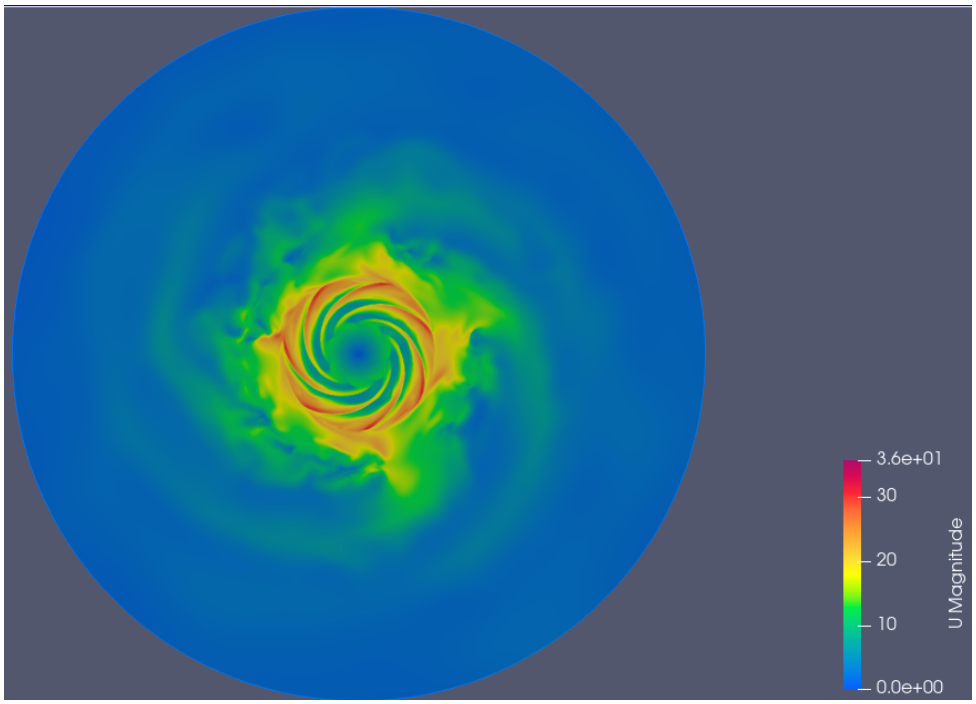


Figure 3.7: Velocity field for Dx = Dy, but with parameters too high, which make the simulation diverge

Strategie 2. $\boldsymbol{D_\omega > 0}$ The strategy is to force $\boldsymbol{D_\omega > 0}$, keeping a variance between each other to allow the parameters to vary.

This strategy has always been applied after the first simulation and always worked well.

Strategie 3. Radial expansion The strategy is to use a radial expansion of the five forcing coefficients as illustrated in the Figure 3.8. We used a second order polynomial to represent the radial expansion of the forcing coefficients, for a few raisons:

- it allows to have a more complete representation of the forcing coefficients, which can help to better fit the experimental data;
- it allows to have a smooth variation of the coefficients over the impeller radius, which is more realistic than a constant value;
- it allows to describe pretty well the geometry in a quadratic way, which is more complete than a sinusoidal representation (see Figure 3.9) which would also require 3 parameters;
- three parameters are not too many so that the optimisation is not too long and the convergence is not too difficult;
- it permits to eliminates a step of the value to 0 at the blade tips for a continuity of order 0, which can be important for the numerical stability of the simulation.

In that case, the parameters vector change from five values to fifteen values.

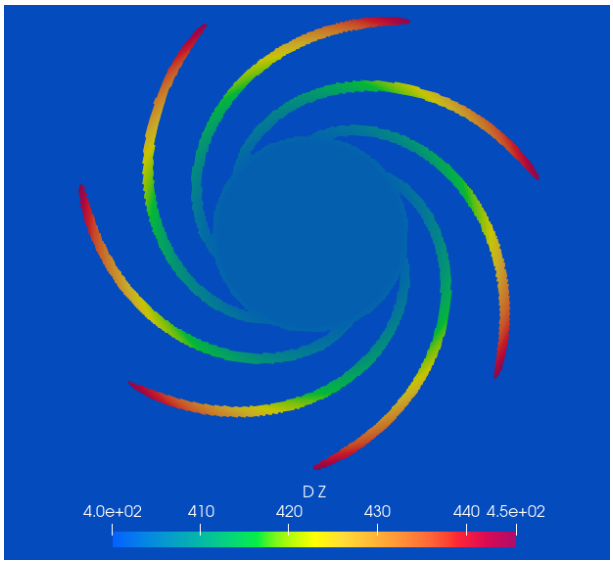


Figure 3.8: Radial expension of Dz over the impeller radius

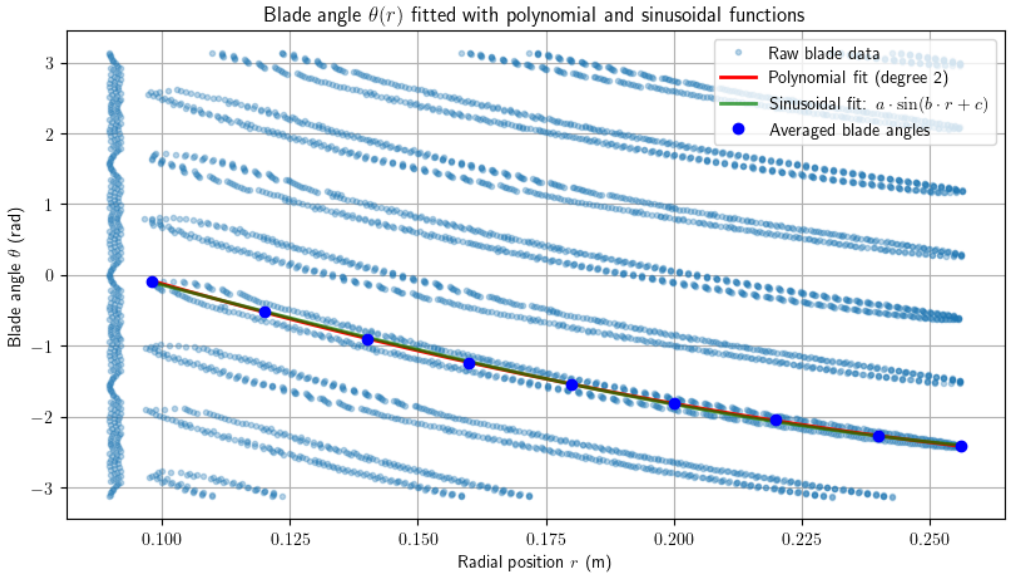


Figure 3.9: Polynomial and sinusoidal interpolation of the blade shape

Strategie 4.Inflation The inflation strategy (explained in the Equation 2.41) is to use a more complex DA method, such as the EnKF with inflation, which allows to escape from local minima by adding noise to the ensemble members. This can help to explore the parameter space more effectively and find a better solution. But we finally didn't have this problem in our case so it hasn't been used.

Strategie 5.Pressure observation Finaly, we can also add a pressure observation to the DA, which allows to take into account the pressure difference between the inlet and the outlet of the pump. This can help to improve the convergence of the solution and have a more physical result. The pressure observation represents the difference of pressure between the inlet of the pipe and the outlet of the diffuser. The numerical coefficients are interpolating, respectively at (0,0,-0.35) and (0,0.390,0). Incertitude on p will be less than the measured one to take it better into account. This implementation add one element on the observation vector.

These strategy are not enough as we can see in the Figure 3.10 where the NRMSEs (computed

between the numerical simulation and the PIV data) remain high and the velocity field is still not physical and not converged. The NRMSE on the highest PIV plan (75%) is significantly reduced. This is locally true for the observation points, but not globally for the whole plan.

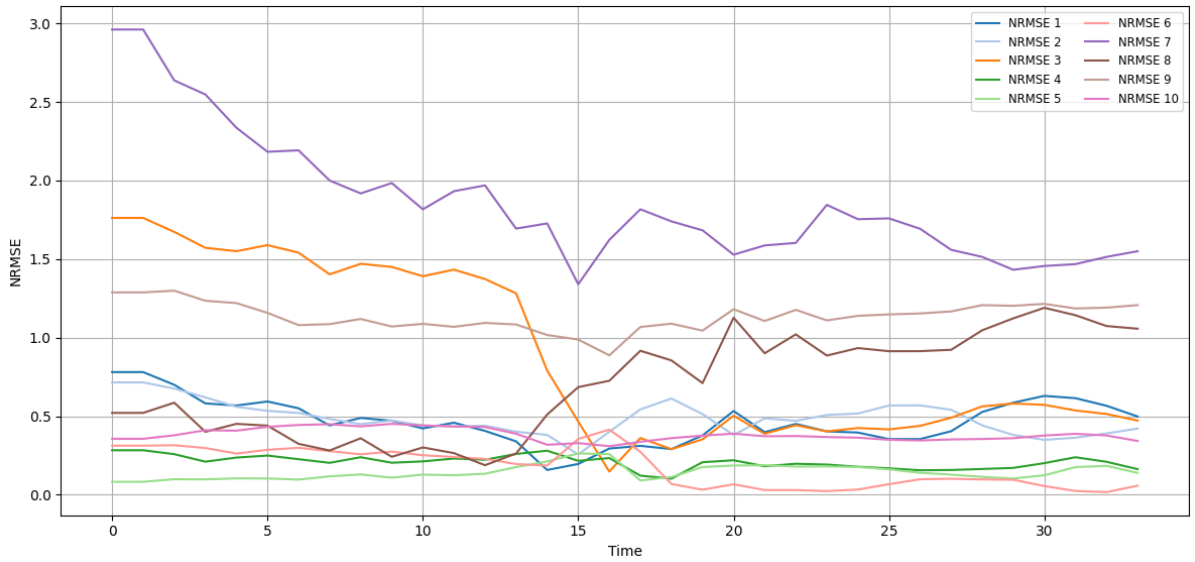


Figure 3.10: Evolution of the NRMSEs over EnKF analysis phases

More precisely, on the Figure 3.10, the legend is organised as follows: NMRSE 1 to 3 are for the radial velocities at respectively 25%, 50% and 75% of the impeller height; NRMSE 4 to 6 for the azimuthal velocities and NRMSE 7 to 9 are oredered following the same ideal; NRMSE 10 is for the pressure difference between the inlet and the outlet of the pump.

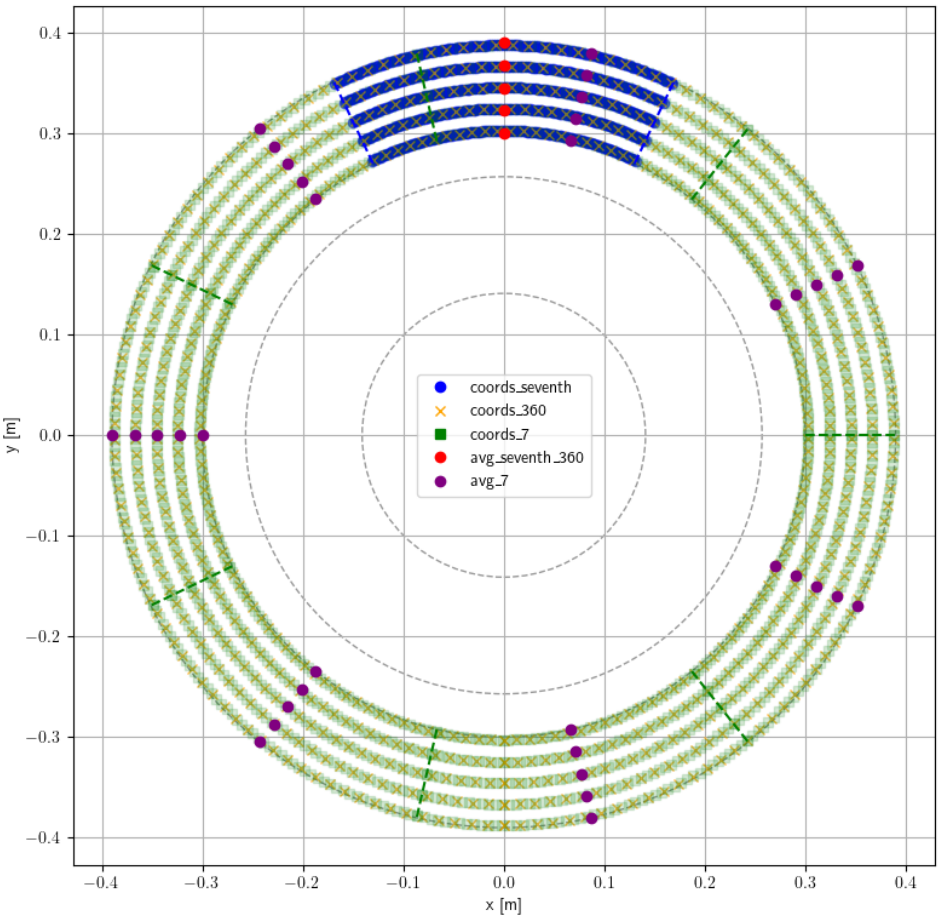


Figure 3.11: Illustration of the augmented coordinates for the interpolation of the field

Strategie 6. More global interpolation

The following explanations are based on the Figure 3.11. The minimum is found based on the blue cercles "coords_seventh" interpolated points and averaged on the "avg_seventh_360" red point localisation. To try to have a more uniform field solution, we can average the parameters over a larger area, i.e., over 360 degrees instead of **360/7** degrees. This allows to have a more global view of the parameters and to avoid local minima (green squares "coords_7" average on the red cercles). But this can also lead to a solution that does not respect the periodicity of the pump, i.e., the solution is not 7 times the same pattern. So we can also average the parameters over 7 segments of **360/7** degrees, which allows to have a more uniform field solution while respecting the periodicity of the pump (orange cruses "coords_360" and purple cercles "avg_7"). In this last case, the length of the coordinate of the observation points is 105 (5 variables, 3 PIV planes, and 7 segments of **360/7** degrees) and the length of the observation vector is 315 (3 dimensions, 105 points).

After applying these strategies, the solution is still not satisfactory, NRMSEs remain high and the velocity field is still not physical and not converged (see Figure 3.12). The NRMSEs remaines high, especially for the radial velocity and velocity over z, because of the no physical recirculation bubble.

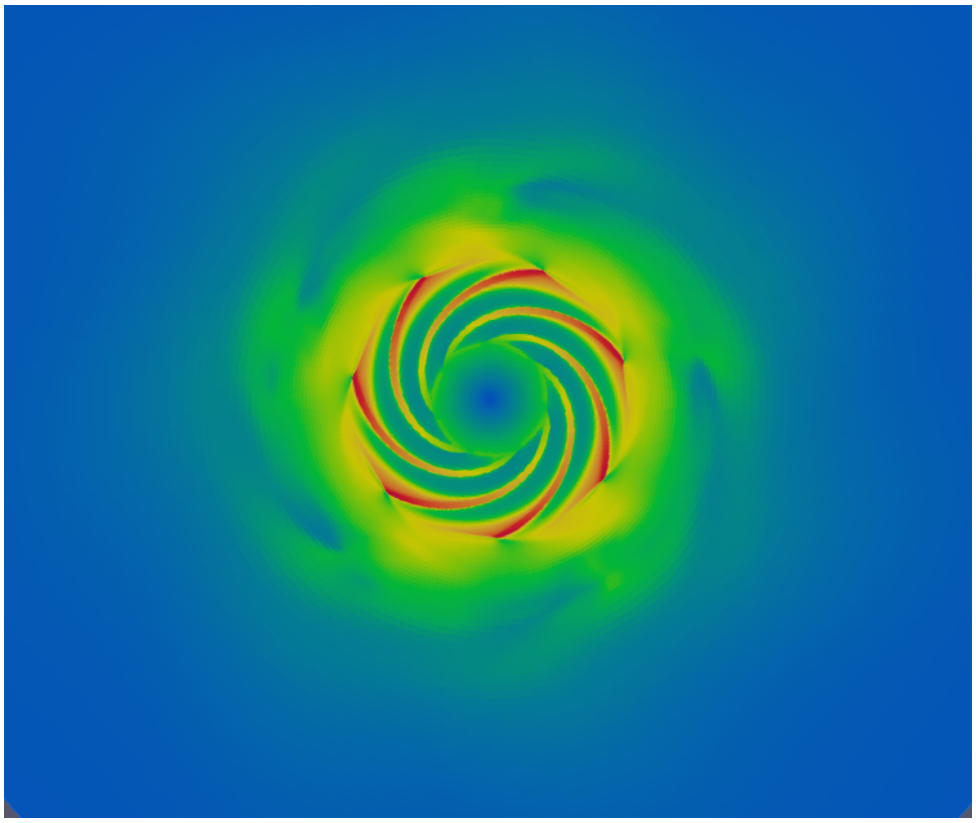


Figure 3.12: Velocity field diverged after applying all the strategies

After applying these strategies, the solution is still not satisfactory, as we can see in the Figure 3.10 where the NRMSEs remain high and the velocity field is still not physical and not converged.

Conclusion

Technical and human review

The objective of this work was to use a DA algorithm to improve an IBM for the simulation of a centrifugal pump. Four milestones were defined: assess the limitations of the baseline IBM, prepare experimental data, integrate DA into the solver, and optimize IBM parameters.

The first technical challenge was to identify the limitations of the baseline IBM. We observed that the method did not correctly impose wall boundary conditions, which generated non-physical recirculation zones in the diffuser. From a human perspective, this stage required learning how to question results critically and not to take numerical outputs at face value. Discussing these anomalies with colleagues helped me develop a more rigorous approach to verification and to clearly communicate uncertainties.

Processing the high-fidelity experimental data (PIV and pressure taps) was another important step. This task was technical, requiring careful handling of noisy measurements, but also human, as it highlighted the importance of collaboration with experimentalists. Their expertise was essential to understand the limits of the data, and I learned to value multidisciplinary exchanges between numerical and experimental teams.

The integration of the DA method via the EnKF into OpenFOAM represented the most demanding technical milestone. It required an in-depth study of OpenFOAM's structure and C++ implementation. Beyond the coding effort, this experience taught me perseverance: many tests failed before achieving a stable coupling. Regular discussions with my supervisors and peers played a key role in finding solutions, reinforcing the collaborative dimension of research.

The final goal, optimizing IBM parameters via DA, could not be achieved. Despite several strategies, the DA corrections did not improve the simulation accuracy and the recirculation bubble persisted. This limitation is not only technical (the IBM formulation itself is insufficient) but also methodological: I learned that in research, negative results are just as important as positive ones. Presenting and accepting such results in a group setting was an exercise in humility and honesty, qualities that are crucial in a research environment.

More generally, this internship allowed me to discover the specificities of an academic laboratory. The freedom to test ideas is great, but it also comes with the responsibility to structure and document one's work to make it useful for others. I imagined similarities with the industrial world, such as social and environmental responsibility, but the emphasis here is placed on knowledge creation.

Finally, this work confirmed the potential of DA as an alternative to expensive LES or DNS for industrial applications, while also underlining the computational and methodological challenges that remain. This balance between technical exploration and human learning convinced me to continue in this field, which led to my decision to pursue a PhD in the same laboratory.

Perspectives

To continue this work, it would be interesting to test alternative IBM formulations, such as the interpolation method, or to run unsteady simulations (URANS or LES) to evaluate whether they can overcome the non-physical results observed with the current approach. The results obtained here suggest that the IBM implementation itself is the main limitation, since no parameter optimization via DA succeeded in removing the strong recirculation bubble in the diffuser.

From a human and methodological perspective, this opens a reflection on how to deal with incomplete or unsatisfactory results. The work showed that adapting DA to complex industrial cases is a long process, with frequent failures and heavy computational requirements. Running DA simulations often meant waiting several days before obtaining results, which required patience, organization and optimization of resources. I also learned that negative outcomes are part of research and can be valuable stepping stones for future improvements.

The DA method still has a strong potential for industrial applications, but it faces challenges: reducing the computational cost, improving convergence, and adapting algorithms to complex geometries and real-time constraints. Exploring hybrid strategies, for instance coupling DA with machine learning techniques, could help overcome some of these barriers. In parallel, improving IBM formulations is essential to better capture wall effects and provide physically consistent flow predictions.

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Appendices

Appendix 1 – Discretization and coupling in OpenFOAM (RANS / U-RANS)

In OpenFOAM (finite-volume), the RANS/U-RANS momentum equation is assembled with an *effective* viscosity $\boldsymbol{\mu_{\text{eff}} = \mu + \mu_t}$ supplied by the turbulence model:

$$\boldsymbol{\rho\left(\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U}\cdot\nabla)\mathbf{U}\right) = -\nabla p + \nabla\cdot\left[\mu_{\text{eff}}\left(\nabla\mathbf{U} + \nabla\mathbf{U}^{T}\right)\right] + \mathbf{f}.} \tag{7}$$

Using the Boussinesq hypothesis one has $\boldsymbol{\mu_t = \rho\,\nu_T(k,\omega)}$ from the $\boldsymbol{k\omega}$ SST model.

Finite-volume form (symbolic):

$$\underbrace{\mathbf{fvm :: ddt}(\rho,\mathbf{U})}_{\text{unsteady}} + \underbrace{\mathbf{fvm :: div}(\rho\,\phi,\mathbf{U})}_{\text{convection}} - \underbrace{\mathbf{fvm :: laplacian}(\mu_{\text{eff}},\mathbf{U})}_{\text{diffusion}} = -\nabla p + \mathbf{f}, \tag{8}$$

where $\boldsymbol{\phi = \mathbf{S}_f \cdot \mathbf{U}_f}$ is the face flux (surface vector $\mathbf{S}_f$ times face velocity), possibly density-weighted for compressible solvers.

Pressurevelocity coupling (PISO/PIMPLE):

(i) Momentum predictor: $$\mathbf{A}_U\,\mathbf{U} = \mathbf{H}_U - \nabla p, \tag{9}$$

(ii) Continuity (incompressible): $$\nabla\cdot\mathbf{U} = 0, \tag{10}$$

(iii) Pressure eqn from flux continuity: $$\nabla\cdot\left(\frac{1}{\mathbf{A}_U}\nabla p\right) = \nabla\cdot\left(\frac{\mathbf{H}_U}{\mathbf{A}_U}\right), \tag{11}$$

(iv) Velocity correction: $$\mathbf{U} \leftarrow \frac{\mathbf{H}_U - \nabla p}{\mathbf{A}_U}. \tag{12}$$

Here $\mathbf{A}_U$ is the diagonal of the discretized momentum operator (from `fvm::ddt`, `fvm::div`, `fvm::laplacian`) and $\mathbf{H}_U$ collects the explicit terms.

Turbulence-model coupling: At each time step (or outer iteration),

solve $\boldsymbol{k}$ and $\boldsymbol{\omega}$ transport eqs. with $\mathbf{U}$, (13)

$\boldsymbol{\nu_T \leftarrow \nu_T(k,\omega)}, \quad \boldsymbol{\mu_{\text{eff}} \leftarrow \mu + \rho\,\nu_T}$, (14)

reassemble momentum operator with $\boldsymbol{\mu_{\text{eff}}}$, then PISO/PIMPLE correct $\boldsymbol{(p,\mathbf{U})}$. (15)

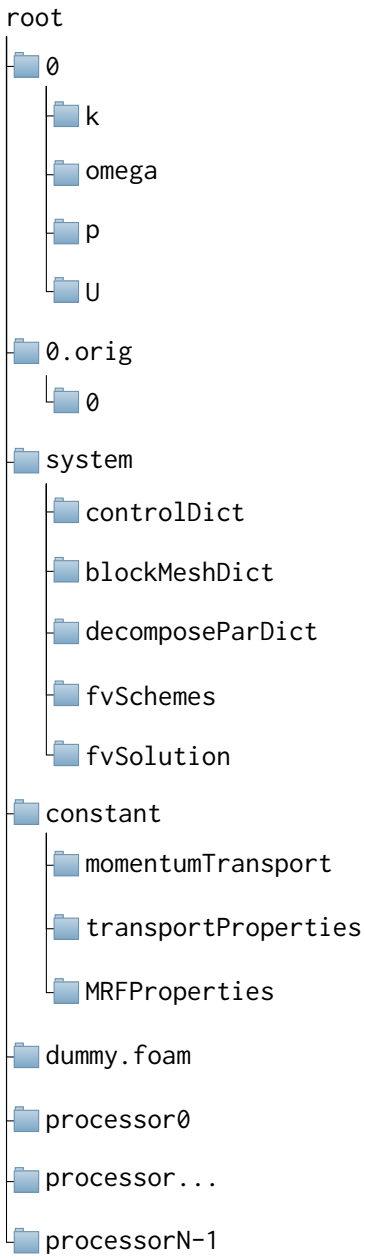
OpenFOAM-style pseudo-code (for clarity):

```
volScalarField nuEff = nu + turbulence->nut();  // mu_eff / rho in incompressible solvers

// UEqn: momentum predictor
fvVectorMatrix UEqn
  (
    fvm::ddt(U)
  + fvm::div(phi, U)
  - fvm::laplacian(nuEff, U)
  == source   // e.g. IBM force, body forces
  );
UEqn.relax();
```

```
// Turbulence update (k-omega SST)
turbulence->correct();          // solves k, omega and updates nut()

// Pressure-velocity coupling
while (pisoOrPimple.loop())
{
    // H and A of momentum equation
    surfaceScalarField rAU = 1.0/UEqn.A();
    U = rAU*UEqn.H();
    phi = fvc::flux(U);          // face flux from predicted U

    // Pressure equation
    while (pisoOrPimple.correct())
    {
        fvScalarMatrix pEqn
        (
          fvm::laplacian(rAU, p) == fvc::div(phi)
        );
        pEqn.solve();

        // Flux and velocity corrections
        phi -= pEqn.flux();
        U -= rAU*fvc::grad(p);
        U.correctBoundaryConditions();
    }
}
```

This makes explicit where the turbulence model enters: via $\boldsymbol{\nu_T}$ in `nuEff`, which modifies the diffusive term in the discretized momentum equation; the $\boldsymbol{k}$ and $\boldsymbol{\omega}$ equations are advanced each step to refresh $\boldsymbol{\nu_T}$ before (or within) the PISO/PIMPLE loop.

Appendix 2 – OpenFOAM Directory Tree



Appendix 3 – Extract of conesDict

Extract from CONES Github[2].

```
paramEstSwitch true; // Switch for parameter estimation
stateEstSwitch false; // Switch for state estimation

// Number of time steps between each analysis phase
observationWindow 800;

// Number of members in the ensemble
ensemble 35;

fineEnsemble 0;
simulationSubDict
{
numberVariablesInState 3;
}

// Number of parameters to optimize with the DA
numberParameters 15;

// Dimensions of the OpenFOAM simulation
nDim 3;

observationSubDict
{
    // Name of the file containing the observation coordinates
    obsCoordFile "obs_coordinates7.txt";

    // Name of the file containing the observation data
    obsDataFile "fieldP7.txt";

    // Number of observation probes for velocity
    numberObsProbesVelocity 105;

    // Specify velocity components in the observation
    obsVelocityComponents (1 1 1);

    // Number of observation probes for pressure
    numberObsProbesPressure 0;

    // Number of observation probes for friction coefficient (default = 1)
    numberOfGlobalObservation 1;

    // pertubation of the obs and diagonal of R matrix for velocity
    velocityObsNoise (10 10 10);

    // pertubation of the obs and diagonal of R matrix for pressure
    pressureObsNoise 0;

    // pertubation of the obs and diagonal of R matrix for the global coefficient
    globalObsNoise 5;

    // Switch to for the obs: true: obs depends on time, false: obs does not depend
        on time
    obsTimeDependency false;

    // The time step of the observations if the case is unsteady
    obsTimeStep 0.004;

    // The start time for the observation data
    obsStartTime 0.0;
```

[2]From:https://gitlab.ensam.eu/pe431/cones-dev/-/blob/pump_clem/tests/pumpIBM/pumpIBMSimple35/system/controlDict

```
59
60      // Def of obs parameter:  U(velocity), p(pressure), Up(velocity + pressure), UCd
            (velocity + drag/friction coeffs), UGlobal(velocity + global obs)
61      obsType "UGlobal";
62
63      // Inputs for R are given in absolute values (absVal), percentage (relVal)
64      obsNoiseType "absVal";
65  }
66
67
68  inflationSubDict
69  {
70      // Definition of inflation (0 = stochastic, 1 = deterministic)
71      inflationType "deterministic";
72
73      // State inflation (default = 0)
74      stateInflation 0;
75
76      // Pre State inflation (considered as model error) (default = 0)
77      preStateInflation 0;
78
79      // Do we apply inflation (default = false)
80      inflationSwitch false;
81
82      // Parameters inflation
83      parametersInflation (0.2 0.2 0.2 0.2 0.2);
84  }
85
86  // Number of txt file written beginning from the last one
87  numberOutputs 1;
88
89  // Print all the debuging messages or not: 1 = printed, 0 = nothing
90  verbosityLevel 2;
```

Appendix 4 – Batch code

```
#!/bin/bash                      # To be run with bash
#MSUB -r LCP35                   # Job name
#MSUB -o LCP35_%I.out            # Output name. %I is the job ID
#MSUB -e LCP35_%I.err            # Error name. %I is the job ID
#MSUB -q rome                    # Partition
#MSUB -A gen1741                 # Project ID
#MSUB -n 4446                    # Cores nb (35 members x 127 OF) + 1 python
##MSUB -c 2                      # To allocate 2 cores per task (default is 1)
##MSUB -N 1                      # Nb of calcul node (128 cores per node on rome)
#MSUB -T  43200                  # Elapsed time limit in seconds
##MSUB -Q long                   # (upto 3 days)
##MSUB -Q test                   # (short, high priority jobs)
#MSUB -m scratch                 # Excecute in scratch directory
#MSUB -@ name@mail.fr:begin,end # Email address

set -x
cd ${BRIDGE_MSUB_PWD}
module purge
module load gnu/8 mpi/openmpi/4 flavor/python3/cuda python3/3.10.6 openfoam/9
source $FOAM_SOURCE_FILE

./visuNParams.sh                 # Script .sh
ccc_mprun -f app_mySF35.conf     # Run the application with the configuration file
```

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List of acronyms

AI Artificial Intelligence

BC Boundary Conditions

CERFACS "Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique"

CFD Computation Fluid Dynamics

CFL Courant-Friedrichs-Lewy

CHSCTN "Comité d'Hygiène, de Sécurité et des Conditions de Travail National"

CNRS "Centre National de la Recherche Scientifique"

CONES Coupling OpenFoam with Numerical EnvironmentS

CPER "Contrat de Plan État-Région"

CPU Central Processing Unit

CSE "Comité Social et Économique"

CSR Corporate Social Responsibility

CTI "Commission des Titres d'Ingénieur"

DA Data Assimilation

EnKF Ensemble Kalman Filter

ENSAM "École Nationale Nupérieure d'Arts et Métiers"

HCERES "Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur"

IBM Immersed Boundary Methods

IPSA "Institut Polytechnique des Sciences Avancées"

JFM Journal of Fluid Mechanics

KF Kalman Filter

LMFL "Laboratoire de Mécanique des Fluides de Lille"

Ma Mach number

ML Machine Learning

MPI Message Passing Interface

MRF Multiple Reference Frame

NRMSE Normalized Root Mean Square Error

ONERA "Office National d'Études et de Recherche Aérospatiales"

OpenFOAM Open source Field Operation And Manipulation

PIV Particle Image Velocimetry

QVT "Qualité de Vie au Travail"

Re Reynolds number

RSE "Responsabilité Sociétale des Entreprises"

RSU "Rapport Social Unique"

SHF "Société Hydrotechnique de France"

SST Shear Stress Transport

TGCC "Très Grand Centre de Calcul du CEA"

URANS Unsteady Reynolds-Averaged Navier-Stokes

Résumé / Abstract

Résumé Ce stage de fin d'études clôture mon cursus à l'IPSA Toulouse. Il a été réalisé au Laboratoire de Mécanique des Fluides de Lille (LMFL), sur le campus de l'Ecole Nationale des Arts et Métiers. En s'appuyant sur les travaux déjà présents dans la littérature, notamment au LMFL, l'objectif du stage intitulé *Optimisation des performances d'une pompe centrifuge à l'aide de l'assimilation de données* était de faire de l'**Assimilation de Données** en utilisant la librairie d'**CONES** pour combiner des simulations **OpenFOAM** avec des données prise sur une **pompe centrifuge** au LMFL. Cela dans l'objectif d'améliorer la simulation numérique utilisant une **méthode aux frontières immergées (IBM)**. La méthode choisie pour l'assimilation des données est le **filtre de Kalman d'ensemble (EnKF)**, qui permet de combiner les données de simulation et les observations expérimentales pour obtenir une estimation plus précise de l'état du système. Les résultats obtenus montrent un défault de la méthode IBM dans la zone du diffuseur, qui n'a pas pu être corrigé par l'assimilation de données. Cependant, l'implémentation de l'EnKF dans CONES a bien été réalisée et des perspectives d'amélioration ont été identifiées pour corriger ce défault. Ce stage ouvre sur une thèse que je commencerai en octobre 2025 au LMFL, sur un sujet similaire.

Mots-clés: **Assimilation de Données, OpenFOAM, CONES, Pompe centrifuge, filtre de Kalman d'ensemble (EnKF), méthode aux frontières immergées (IBM)**

Abstract This end-of-studies internship finalises my curriculum at IPSA Toulouse. It was carried out at the Fluid Mechanics Laboratory of Lille (LMFL), on the campus of the Ecole Nationale des Arts et Métiers. Building on previous work in the literature, notably at LMFL, the objective of the internship entitled *Optimization of the performance of a centrifugal pump using Data Assimilation* was to perform **Data Assimilation (DA)** using the **CONES** library to combine **OpenFOAM** simulations with data taken on a **centrifugal pump** at LMFL. The goal was to improve the numerical simulation using an **Immersed Boundary Bethod (IBM)**. The chosen method for data assimilation is the **Ensemble Kalman Filter (EnKF)**, which allows to combine simulation data and experimental observations to obtain a more accurate estimate of the state of the system. The results obtained show a flaw in the IBM method in the diffuser area, which could not be corrected by data assimilation. However, the implementation of the EnKF in CONES was successfully carried out and improvement perspectives were identified to correct this flaw. This internship opens the way to a PhD that I will start in October 2025 at LMFL, on a similar subject.

Keywords: **Data Assimilation (DA), OpenFOAM, CONES, Centrifugal pump, Ensemble Kalman Filter (EnKF), Immersed Boundary Bethod (IBM)**